

relatively high light intensity. As the forest canopy absorbs more red light allowing only far red light to pass through, the tree below the canopy will allocate more of its resources to grow taller.

- In contrast, exposure to direct sunlight increases the proportion of far red: red light and thereby stimulates branching and inhibits vertical growth.

flowering

- Photoperiod is the interval in a 24hour period in which the plant gets exposed to light.
- Photoperiod controls flowering in many types of plants.

Shoot elongation and Phototropism

- The growth of a shoot towards light (positive) or away from it (negative) is called phototropism.
- Positive phototropism strengthen photosynthesis.
- This response results from a differential growth of cells on opposite sides of the shoot; the cells in the darker side elongate faster than the cells on the brighter side.

Response to Gravity

Gravitropism

- Shoot of the plant grows upwards while root grows downwards, due to their response to gravity or gravitropism.
- Gravitropism can be either positive or negative.
e.g. Roots display positive gravitropism while shoot display negative gravitropism.
- Gravitropism occurs as soon as a seed germinates. This ensure that the root grows into the soil and shoot grows towards sunlight.
- Plants may detect gravity by the settling of statoliths.
- Statoliths of vascular plants are specialized plastids containing dense starch grains.
- They can settle under the gravity, to the lower portions of the cell.
- In roots, they are located in certain cells of the root cap.

The statolith hypothesis:

The aggregation of statoliths at the low points of root cap cells triggers re-distribution of Ca^{2+} which causes lateral transport of auxin within the root. As a result, Ca and auxin get accumulated at lower side of elongation zone of root. At high concentration of auxin, cell elongation is inhibited resulting slow growth on lower side and more rapid elongation on upper side. Consequently, the root grows downwards.

Response to mechanical stimuli

Trees grow in windy environment normally have shorter stockier trunks than same species growing in normal environmental conditions. Advantage of this is that the tree could stand high winds. This exhibits the sensitivity of mechanical stress of plants. The changes in plant form due to mechanical disturbances is called thigmomorphogenesis. During evolution, some plant species have become 'touch specialists'. Climbing plants have tendrils that coil rapidly around support. Tendril usually grows straight until it touches a support. The contact stimulates differential growth on opposite sides of the tendril. The directional growth of tendril towards support is called thigmotropism.

Other touch specialists, respond to touch by rapid leaf movements. E.g. *Mimosa pudica* collapses its leaflets when touched. Touching results in a sudden loss of turgor of cells in a specialized motor organ called pulvini, causing the leaflets to collapse. This response is called thigmonasty.

The role of plant growth substances/ hormones/ regulators in response to different stimuli

Hormones in general are signaling molecules which are produced in small quantities, get transported from the place they are produced to other parts of the organism and trigger responses in target cells. or/and effect on plant growth and development. With this definition, its hard to explain some physiological processes in plants. In addition, some signaling molecules that are considered as plant hormones act locally. Thus the broader term plant growth regulators seem more appropriate.

Plant growth regulators are natural or synthetic organic compounds which modify or control specific physiological process in plants.

Plant biologists prefer to use the term plant growth regulators rather than plant hormones, as there are certain differences in plant hormones and animal hormones.

Therefore, plant hormones and plant growth substances are considered as equal. But plant hormones are active even at very low concentration.

Major types of plant hormones/ growth regulators are auxins, gibberlin, cytokinin, abscisic acid, ethylene and Jasmonate (jasmonic acid).

Hormone	Functions
Auxin	- stimulates stem elongation in low concentration - promotes the formation of lateral and adventitious roots - regulates development of fruit - enhances apical dominance - functions in phototropism - functions in gravitotropism - promotes vascular differentiation - retards leaf abscission
Gibberellins	- stimulate stem elongation - stimulate pollen development - stimulate pollen tube growth - stimulate fruit growth - stimulate seed development and germination - regulate sex determination and transition from juvenile to adult phase
cytokinins	- regulate cell division in shoots and roots - modify apical dominance and promote lateral bud growth - promote movement of nutrients into sink tissues - stimulate seed germination - delay leaf senescence
Abscissic acid	- inhibits growth - promotes stomatal closure during drought stress - promotes seed dormancy and inhibits early germination - promotes leaf senescence - promote desiccation tolerance
Ethylene	- promotes ripening of many types of fruit - promote leaf abscission - promote triple response in seedlings (inhibition of stem elongation, promotion of lateral expansion, and horizontal growth) - enhance the rate of senescence - promote roots and root hair formation - promotes flowering in the pineapple family

Response of plants to some biotic and abiotic stresses

Stress

Certain factors in the environment may have a potentially adverse effects on a plants' survival, growth and reproduction.

Two types of stresses,

1. Abiotic stress
2. Biotic stress

Response of plants to some biotic and abiotic stresses

Stress

Certain factors in the environment may have potentially adverse effects on plants' survival, growth and reproduction.

Two types of stresses;

3. Abiotic stress(due to nonliving factors)
4. Biotic stress (due to living factors)

Abiotic Stress

Among several common abiotic stresses. Following three stresses are discussed.

1. Drought stress
2. Cold stress
3. Salt stress

1. Drought stress: Plants may wilt when water loss by transpiration exceeds water absorption. Prolonged drought may even kill a plant. Plants have control systems that enable them to cope with the drought/ water deficit conditions.

Water deficit stimulates increased synthesis and release of abscisic acid (ABA), which acts on guard cell membrane, closing stomata to reduce transpiration.

In grasses the leaves roll in to a tube-like shape which reduces the surface area to reduce transpiration. Some plants shed their leaves during seasonal drought.

2.Cold stress: When cell membrane cools below a critical temperature it loses its fluidity due to the lipids become locked in to crystalline structure. This blocks the transport across the membrane and affects the function of the cell.

Plants respond to cold stress by altering the lipid composition of their membranes. They increase the proportion of unsaturated fatty acids which keeps the membranes more fluid at low temperature.

Freezing is another cold stress. Water in the cell wall and intercellular spaces freezes before freezing the solute-rich water in the cytosol. The reduction of liquid water in the cell wall lowers the extracellular water potential causing water in the cytosol to leave.

This results high concentration of solutes in the cytoplasm which is harmful and may lead to cell death.

Before the onset of winter, the cell of frost-tolerant plants increases cytoplasmic levels of specific solutes such as sugars that help to reduce the loss of water from the cell preventing dehydration.

3. Salt stress: An excess of salts (high salinity) in soil lowers the water potential of soil resulting reduced water potential gradient from soil to root. This leads to reduction of water uptake by roots.

In general too high salinity in soil is toxic to plants.

Many plants can respond to moderate soil salinity by producing solutes that are well tolerated at high concentrations. These are organic compounds that keep the water potential of cell more negative than that of the soil solution.

A few plants that are salt-tolerant (halophytes) have developed salt glands, which secrete excess salts out of the plant across leaf surfaces. e.g. many mangrove plants

Biotic stress –

How plants defend themselves against pest and pathogens attack;

In plant defense mechanisms, some compounds and structures are already existed whilst some others are formed after infection or pest attack. Therefore, two categories of defense mechanisms called preexisting and induced mechanisms can be identified.

Preexisting structural and chemical defense mechanisms;

- Amount and quality of wax and cuticle that cover the epidermal cells
- The structure of the epidermal cell walls and thickness
- The size, location and shapes of stomata
- Toxic compounds, alkaloids (eg. Nicotine), phenolics (eg. Flavonoids, lignin & tannins), terpenoids (eg. Azadirachtin) and lectin
- Thorns, pricks, trichomes

Induced structural and chemical defense mechanisms;

- Morphological changes in the cell wall
- Formation of cork and abscission layers
- Phenolic compounds
- Toxic compounds
- Enzymes that can degrade fungal cell walls or damage insect organs

05**Animal form and function****Relate the Structure of Animal Tissues to their function**

There are four major types of animal tissues : Epithelial tissue, Connective tissue, Muscle tissue and Nervous tissue.

1. Epithelial tissue: -**Characteristics**

It covers external or internal free surfaces and organs. The cells of the tissues are closely packed. Cells of epithelial tissue have both apical and basal surfaces where apical surface remains free and the basal surface attached to the basement membrane. No blood vessels in the tissue. Tissue gets nutrients and oxygen from the underneath connective tissue.

Functions-

- Protection (e.g. barrier against mechanical injury, pathogens, barrier against fluid loss)
- Secretion (e.g. enzymes, hormones, mucus, sweat)
- Absorption (e.g. nutrients, respiratory gases)

There are two general types according to the number of cell layers on the basement membrane.

- Simple epithelia – Single cell layer (e.g simple squamous, simple cuboidal, simple columnar and pseudostratified)
- Compound epithelia – several cell layers (e.g. stratified squamous, transitional)

Simple squamous epithelium –

It is a single layer with plate like cells. This type of epithelium is thin and leaky. They are found in places where materials exchange by diffusion.

e.g: blood capillaries, alveoli,

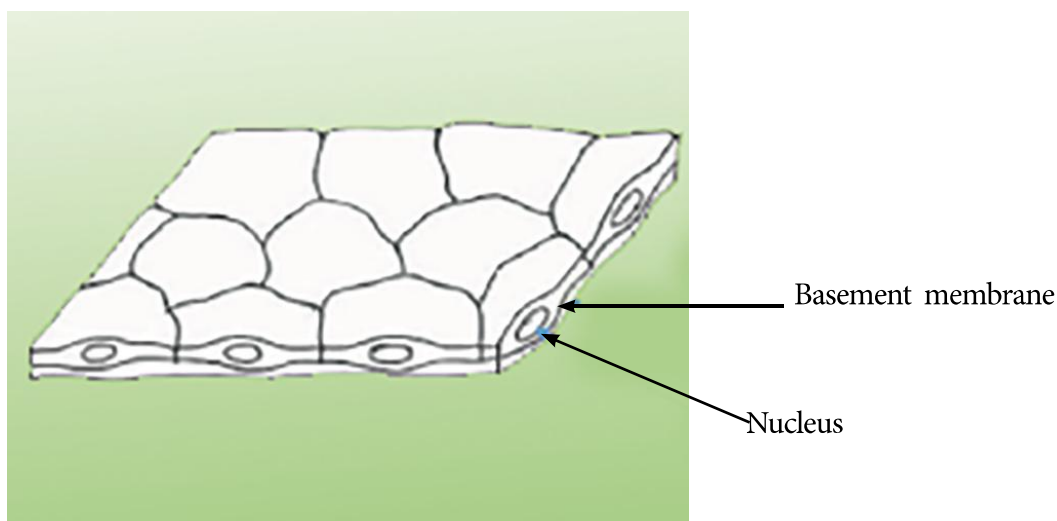


Fig 5.1: simple squamous epithelium

Simple cuboidal epithelium –

It is a single cell layer with dice shaped cells specialized for secretion. It is found in kidney tubules, many glands such as thyroid glands and salivary glands

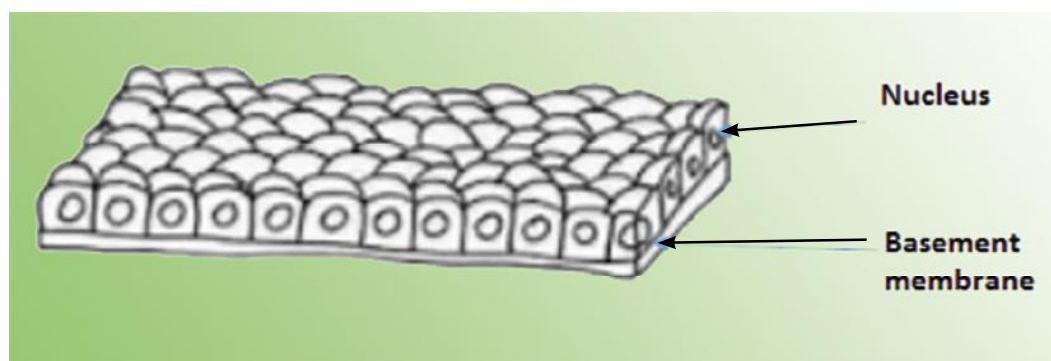


Fig 5.2: simple cuboidal epithelium

Simple columnar epithelium

It is a single layer with large and brick-shaped cells. It is often found in places where secretion or active absorption is important. e.g. intestinal lining

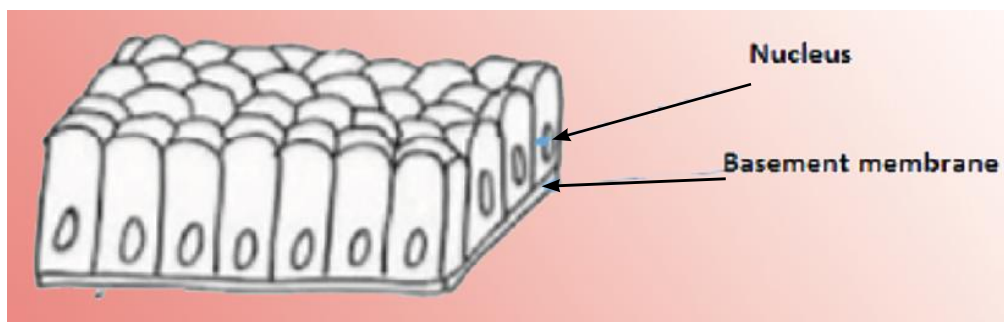


Fig 5.3 : simple columnar epithelium

Pseudostratified columnar epithelium

This tissue consists of a single cell layer. Cells are not of equal height. Nuclei of cells are located at different level. Appear as several layers. In many vertebrates this epithelium has ciliated cells that form a mucous membrane and the cilia help to sweep the mucous along the surface. e.g. nasal passage, trachea.

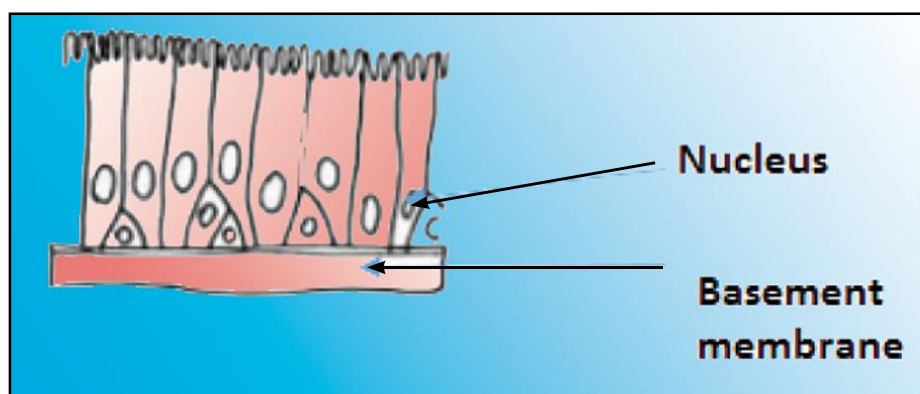


Fig 5.4 : Pseudostratified squamous epithelium

Compound epithelia –**Stratified squamous epithelium:**

This tissue is composed of a number of layers of cells. This epithelium regenerates rapidly. Cell division produces new cells near the basement membrane. The old cells are sloughed off and replaced by the new cells. This epithelium is found on surfaces where they are subjected to abrasion such as outer skin, lining of mouth, anus, vagina

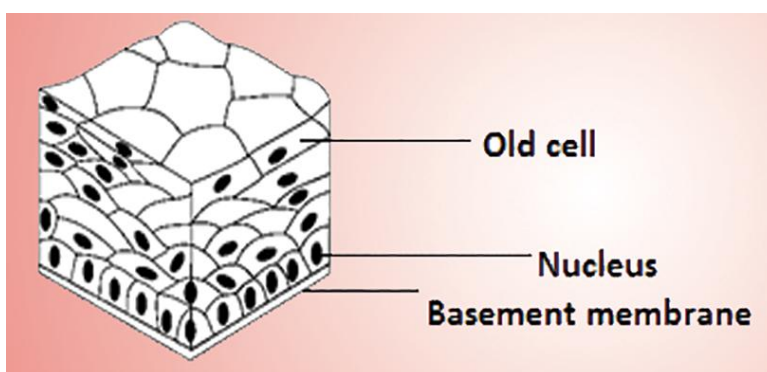


Fig 5.5: Stratified squamous epithelium

2. Connective tissue**Characteristics**

Connective tissues are the most abundant tissues in the body that help to connect organs and tissues together structurally and functionally. These types of tissue consist of different types of cells scattered in a large amount of extracellular matrix containing different types of fibres. Matrix may be a semisolid (Jelly-like), liquid or

solid (dense and rigid). Different types of cells can be found in the matrix such as fibroblasts (secrete fiber proteins), macrophages (engulf foreign particles or any cell debris by phagocytosis) and mast cells (secrete heparin and histamine) in addition fat cells (storage and insulation) and leukocytes (protection) are found in some connective tissues.

There are three different types of fibers. They are collagen fibers (provide strength and flexibility), reticular fibers (join connective tissues to adjacent tissues) and elastic fibers (make tissue elastic).

Functions

- Binding and structural support
- Protection
- Transport of materials
- Insulation

There are different types of connective tissues. They are loose connective tissue (Areolar tissue), Fibrous connective tissue (Dense connective tissue), adipose tissue, blood, cartilage and bone .

Loose connective tissue (Areolar Tissue)

This tissue is the most widely distributed connective tissue type in the vertebrate body. This can be considered as the generalized type of connective tissue. Types of cells in this tissue are fibroblasts, macrophages, mast cells, leukocytes and fat cells.

All three types of fibres are found in this tissue. Fibers are loosely arranged and wavy in nature. This tissue binds epithelia and the underlying tissue. Therefore this holds organs in place. This tissue is found under the skin and throughout the body.

Fibrous connective tissue (Dense connective tissue)

This tissue is densely packed with collagen fibers. Therefore the matrix is relatively reduced and contains fewer cells (fibrocytes). This tissue is found in tendons (attach muscle to bones) and ligaments (connect bones and joints) where tensile strength is required.

Adipose tissue

This tissue is packed with adipose cells. Each adipose cell contains a large fat droplet. This is a specialized type of loose connective tissue which pads and insulates the body and stores fuel as fat molecules. For instance, it is found under the skin where it act as a thermal insulator and energy store.

Blood tissue

It is a specialized connective tissue where the matrix is not secreted by the cells and fibers are formed only during blood clotting process. The extracellular matrix of the blood is liquid. It is called as plasma. The blood plasma contains salts, water and dissolved proteins. Red blood cells (transport respiratory gases), white blood cells (defense) and platelets (blood clotting) are suspended in the blood plasma. Main functions of blood tissues include transport of materials, protection, and osmoregulation. (For further details refer pg 195)

Cartilage

This tissue consists of a matrix composed of chondroitin sulphate which is a rubbery protein-carbohydrate complex. Collagen fibers and chondrocytes are embedded in the matrix. Chondrocytes secrete chondroitin sulphate and collagen fibers. This tissue provides support and flexibility in places such as trachea, intervertebral discs.

Bone

It is a mineralized connective tissue. Matrix consists of collagen fibres and inorganic salts. Inorganic salts are Calcium, magnesium and phosphate ions. Cells are osteoblasts (bone forming cells) and osteocytes (mature bone cells that maintain bone tissue). Osteocytes are enclosed within lacunae. The mammalian hard bone has repeating units called osteons. Each osteons has a concentric layer of mineralized materials. At the centre of the osteon is a central canal containing blood vessels and nerves. This tissue forms the endoskeleton of most vertebrates and provides support and strength to the body.

3. Muscle tissue

Muscle tissue is responsible for movement. The cells in the muscle tissue are composed of actin and myosin proteins. This tissue is able to contract and relax. There are three basic types of muscle tissues found in the vertebrate animal body. They are smooth muscle, skeletal muscle and cardiac muscle tissues.

Smooth muscle tissue

The cells of the smooth muscle tissue are spindle shaped and uninucleated. Cells lack striations. This tissue is responsible for involuntary body functions (e.g. churning of stomach, constriction of arteries). This tissue is found in digestive tract, urinary bladder, arteries and other internal organs.

Skeletal muscle tissue

They are composed of bundles of long cells with multi nuclei. Cells are striated. The contractile units of this muscle cells are called sarcomers. This gives the striated appearance to the muscle cells. The arrangement of the sarcomere gives the striated appearance. The muscles are generally attached to the skeletal system and helps mainly in voluntary body movements.

Cardiac muscle tissue

They are composed of uninucleated cells which are interconnected via intercalated discs. Cells are striated with sarcomeres. Cardiac muscle tissue is responsible for involuntary heart contractions. Intercalated discs help relay signals from cell to cell and synchronize heart contraction. Cardiac muscle tissue is only found in the wall of the heart.

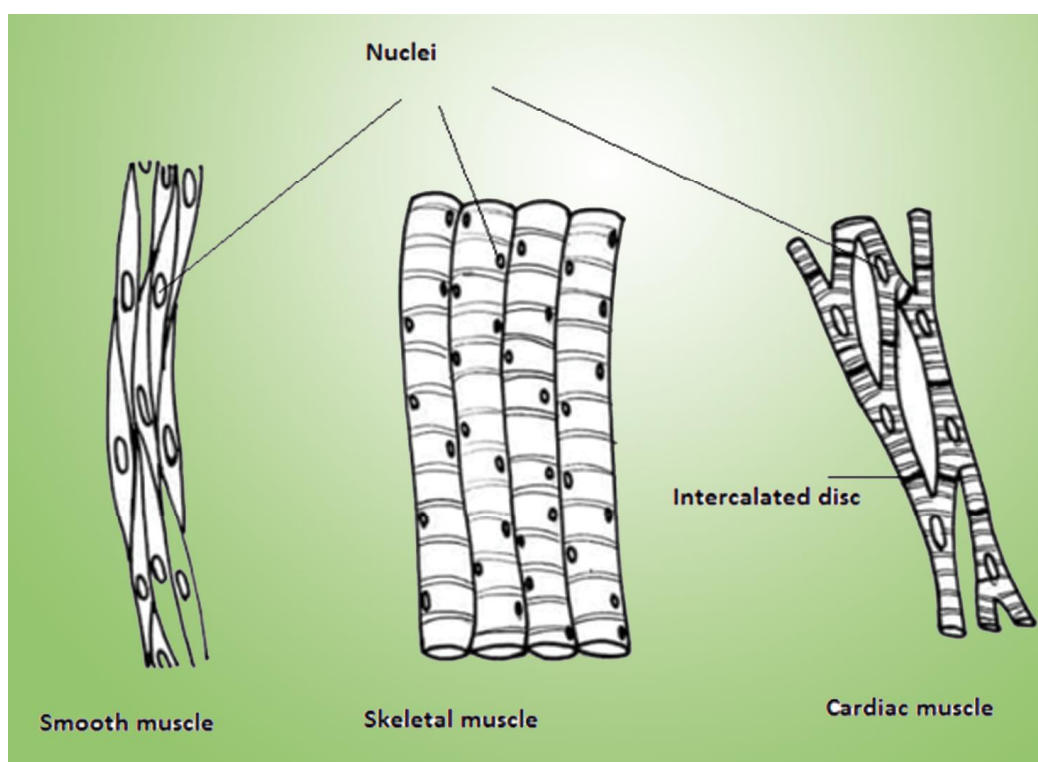


Fig 5.6: Muscle tissue

4. Nervous tissue

Nervous tissue contains neurons and glial cells. Neurons receive, process and transmit nerve impulses. Neuroglia (Glial cells) support the neurons.

Neurons: A neuron has cell body, dendrites and axon. The basic structural unit of the nervous system is the neuron. Dendrites and cell body is used to receive nerve impulses from other neurons. Axon is used to transmit impulses to other neurons, cells or muscles. Axons are bundled together into nerves.

Neuroglia (Glial cells): Neuroglial cells are supportive cells of neurons. Their functions include nourishment of nerve cells, insulation of nerve cells, replenishing neurons and sometimes modulate neuron functions.

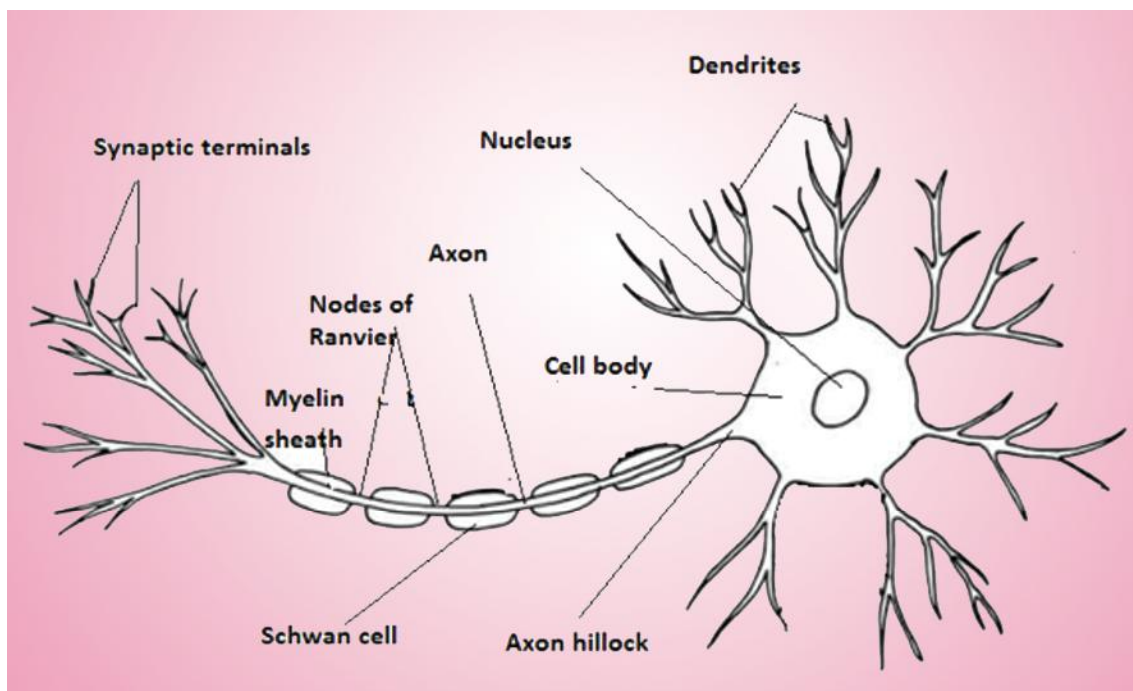


Fig 5.7: The structure of the Motor neuron

Nutrition in animals

Animal nutrition is the process by which food is taken in for make use in different body functions. Typically food is broken into smaller molecules and absorbed prior to utilization. Animals are heterotrophs.

Heterotrophic nutrition is the process by which the organisms obtain organic food molecules by ingesting other organisms or by substances derived from other organisms. Other than animals, fungi and majority of bacteria are heterotrophs. There are two type of heterotrophic nutrition namely, holozoic nutrition and symbiosis.

Holozoic nutrition

Most animals are holozoic where they ingest food into their alimentary canal. This mode of nutrition consists of five main stages: Ingestion, digestion, absorption, assimilation and elimination/ Egestion.

Main stages in holozoic nutrition

- **Ingestion:** This is the first stage where the act of eating or feeding happens. Food sources differ among animal species and they possess different modes of ingestion according to the diet or environment.
- **Digestion:** Food is broken down into molecules small enough to pass through the membranes and enter the cells of organisms. Digestion could occur mechanically (by teeth or muscle contractions) and chemically (by enzymes). During mechanical digestion, food is broken down in to smaller fragments thus increase the surface area for efficient chemical digestion. During chemical digestion enzymes break bonds in large molecules into small molecules.

The last two stages of food processing occur after the food is digested.

- **Absorption:** In this stage, the animal's cells take up small molecules. e.g. simple sugars, amino acids
- **Assimilation:** Assimilation is the process of utilization of absorbed nutrients for various functions of the body.
- **Elimination:** In this process undigested materials are passed out from the alimentary canal.

Feeding Mechanisms of animals

- **Filter feeders:** They strain suspended food particles from the surrounding watery medium. They use several mechanisms such as capturing, trapping, etc. e.g. - clams and oysters - feed on small pieces of food in the water that passes over their gills. Cilia in the gills sweep the food particles to the animal's mouth in a film of mucus.

- **Fluid feeders:** They suck nutrient rich fluid from a living host using well adapted mouth parts.
e.g: Mosquitoes-suck human blood, Aphids- suck phloem sap of plants, Bees and humming birds suck -honey from flowers
- **Substrate feeders:** These animals live inside their food source or on the food source eating its way through the food.
e.g. The leaf miner caterpillar - eating through the soft tissues of a leaf
Maggots (fly larvae) - burrow into animal carcasses
- **Bulk feeders:** Animals which eat comparatively large pieces of food. These animals have different types of adaptations to tear the food or to capture the pray (e.g. jaws, teeth, tentacles, claws, poisonous fangs).
e.g. - Most animals including human

- **Symbiosis**
This is an ecological relationship between organisms of two different species that live closely together. It is divided into three groups such as Mutualism, Parasitism and Commensalism

- **Mutualism:** It is a close association between two living organisms of different species which benefits both partners.
e. g. - Cellulose digesting microorganisms in Ruminants and Termites

- **Parasitism:** It is a close association between two living organisms of different species which is beneficial to one (parasite) and harmful to the other (host). Parasites live either within or on the host and derive its nourishment
e.g. –Tape worm and humans, Lice and humans

- **Commensalism:** It is a close association between two living organisms of different species which is beneficial to one and does not affect the other (neither harmful or beneficial)
e.g.- Barnacles attached to whales

Human Digestive system

Structure and function of the human digestive system

Human alimentary canal is a long tube/ tract which connects with external environment and can fulfill steps of holozoic mode of nutrition. Human digestive system consists of alimentary canal and associated glands. The alimentary canal consists of the following parts: oral cavity, pharynx, esophagus, stomach, small intestine, large intestine, rectum and anus. The associated glands include salivary glands, pancreas and liver.

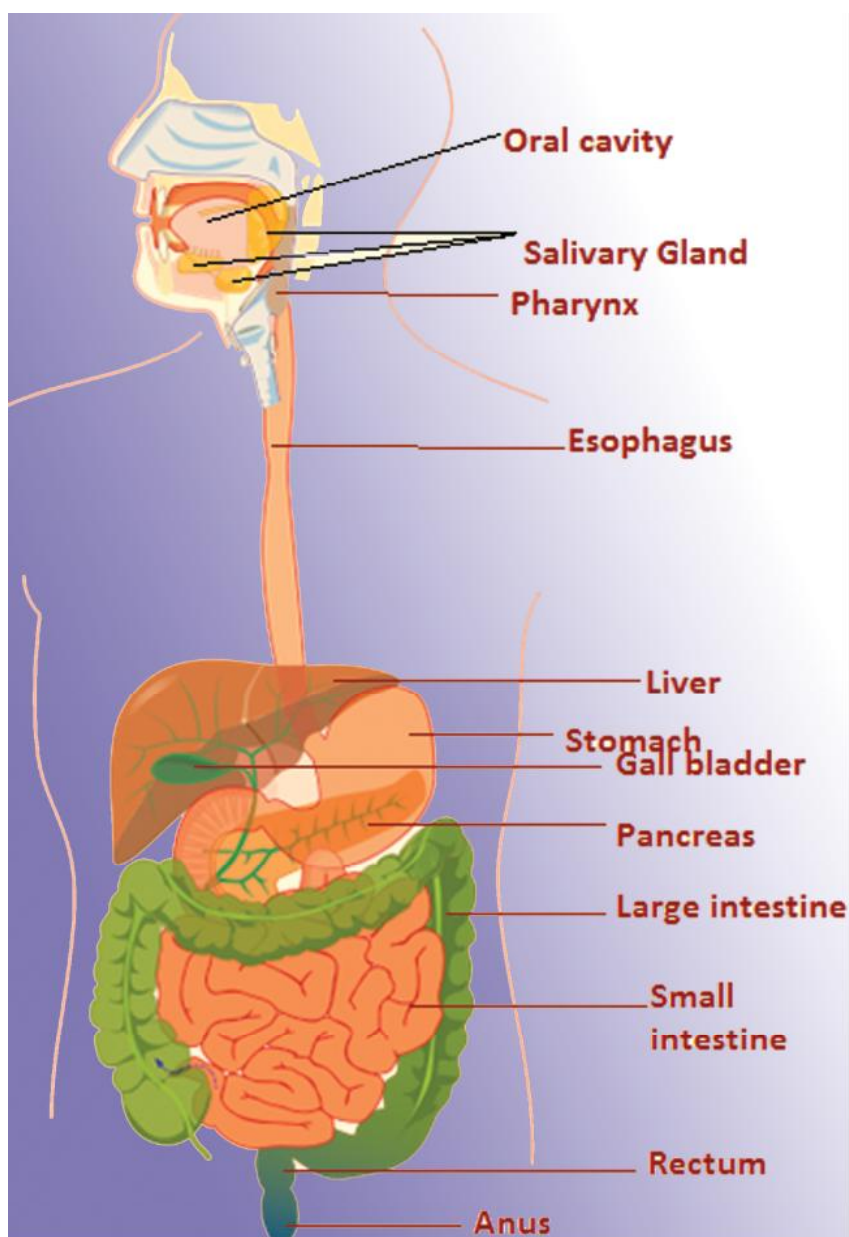


Fig 5.8 :- Structure of the Human Digestive tract

Mouth/ Oral cavity: Oral cavity consists of the tongue, teeth and salivary glands. Ingestion and initial steps of digestion are carried out in the oral cavity. In the mouth both mechanical and chemical digestion is carried out. There are four types of teeth in the mouth: incisors, canines, premolars and molars. Food is cut, mashed and ground by different types of teeth with different shapes. This makes it easier to swallow food and increase the surface area for digestion.

Salivary glands release saliva into the oral cavity through ducts. Releasing saliva into the oral cavity occurs when food enters the oral cavity due to a nervous reflex. Saliva is also released into the mouth before food is ingested due to various other stimuli. e.g. sight, odor of food, etc.

Saliva contains water, amylase, mucus (a viscous mixture of salts, cells and slippery glycoprotein called mucins). In addition to that, saliva composed of buffers and anti microbial components.

Functions of the saliva

- Salivary amylase: Chemical digestion of polysaccharides (e.g. starch) into smaller polysaccharides and disaccharide maltose.
- Water: Liquefy food and provide watery medium for chemical digestion. Aids in taste reception.
- Mucus: Lubrication of food which makes it easier for swallowing. Clean the mouth and protects the lining of the mouth from abrasion.
- Antimicrobial substances such as immunoglobulin and lysozymes: Protect against bacteria that enter the mouth.
- Buffers prevent tooth decay by neutralizing acid

Tongue: composed of skeletal muscles. Helps to mix the food with saliva and make bolus of food that makes easier for swallowing. Then helps to push the bolus into the posterior part of the oral cavity and into the pharynx.

Pharynx: a common passage of the respiratory tract and the digestive tract. The pharynx leads to the esophagus.

Esophagus:

It is a long tube which connects the pharynx and the stomach. It is found in the thoracic cavity. The wall of the esophagus consists of both skeletal and smooth muscles. The skeletal muscles are located towards the uppermost part of the esophagus and these muscles function during swallowing. The rest of the esophagus consists of smooth muscles which are involved in the process called peristalsis. During this process the food bolus is pushed along the esophagus which is a alternative wave of rhythmic contractions and relaxations of the smooth muscles lining the esophagus.

Stomach- The stomach is a J- shaped dilated sac in the abdominal cavity. The inner surface of the stomach is highly folded and contains large number of pits that leads to gastric glands. Gastric glands contain three types of cells: mucus cells, chief cells and parietal cells. The stomach wall is very elastic. Distal part of the stomach connects with the small intestine. Sphincters are found at the junctions between esophagus and stomach (cardiac sphincter) and stomach and the small intestine(pyloric sphincter). They are made up of circular smooth muscles. These sphincters help to regulate the passage of materials between these organs.

Chemical digestion in the stomach

The gastric glands of the stomach secrete gastric juice. The gastric juice mainly consists of mucus, pepsinogen and HCl. Mucus and pepsinogen (inactive form of pepsin) are secreted by mucous cells and chief cells respectively. The parietal cells release hydrogen ions and chloride ions separately into the stomach lumen where HCl is formed. Pepsinogen is initially converted into pepsin by HCl. These activated pepsin help to activate remaining pepsinogen molecules. This activated pepsin initiates the chemical digestion of proteins in the stomach. The churning action of the stomach facilitates the chemical digestion. This is a series of muscle contraction and relaxation. This process mixes the swallowed food with gastric juice. Proteins are hydrolyzed to small polypeptides by pepsin. In the stomach food is mixed with gastric juice forming chyme (partially digested semisolid, acidic, food mass).

The stomach lining is protected from the digestion of HCl and pepsin in several ways: Enzymes are secreted in to the lumen as an inactive enzyme; Gastric glands secrete mucus that protect against self-digestion of the stomach lining ; Every three days, cell division adds a new epithelial cell layer which replaces the destroyed/ damaged cells in the lining of the stomach.

Functions of the stomach

- Act as a temporary reservoir for food due to high convolution and very elastic wall.
- Mechanical digestion of food by churning action due to muscular contraction.
- Produce gastric juice which starts the chemical digestion of proteins to polypeptide by pepsin
- Absorption of some materials such as water, alcohol and some drugs
- Non specific defense-HCl kills microbes
- Small jets of gastric contents push out through pyloric sphincter as chyme
- Secretion of gastrin hormone which regulates digestion in the stomach.

Small intestine-

It is the longest organ in the alimentary canal. It is divided into three regions: duodenum, jejunum and ileum. The duodenum: C shaped curve, around the head of the pancreas. The jejunum is middle part of the small intestine. The ileum is the terminal part of the small intestine. The surface area of the small intestine is greatly increased by permanent circular folds and villi. The villi are tiny finger like projections of the intestinal wall. Most of the digestion is completed in the duodenum. The major sites for nutrient absorption are jejunum and ileum.

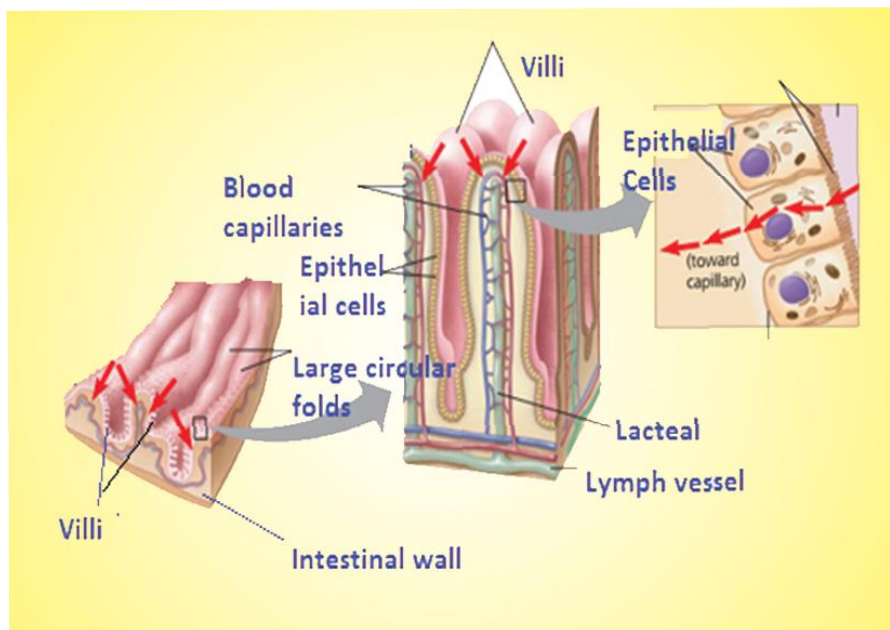


Fig 5.9 Structure of the villus

Chemical digestion in the small intestine

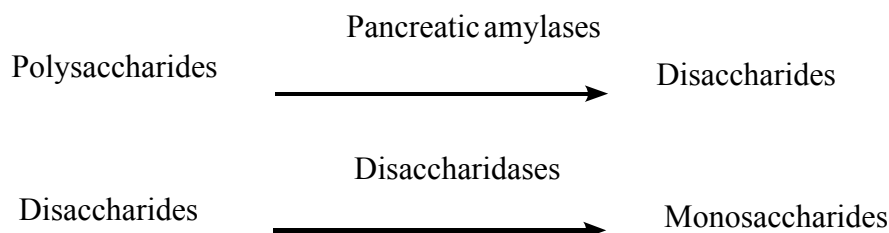
Small intestine receives chyme from the stomach. Peristaltic contractions aid the receiving of this chyme. Pyloric sphincter regulate the passing of chyme into the small intestine. The chyme is mixed with secretions of glands in intestinal wall and secretions of pancreas and liver. (refer pages no: 169,170). The epithelium of the duodenum secretes several digestive enzymes. Glands of the intestinal wall secrete enzymes such as Disaccharidases, Dipeptidases, Carboxypeptidases, Aminopeptidases, Nucleotidases, Nucleosidases and Phosphotases. Some of these enzymes are secreted to the lumen and others are bound to the surface of the epithelium.

Two hormones namely cholecystokonin and secretin secreted by duodenum stimulates the release of pancreatic juice and the bile. Pancreatic juice contains enzymes such as Trypsin, Chymotrypsin, Pancreatic amylase, Pancreatic Carboxypeptidases, Pancreatic Nucleases and Pancreatic Lipases. In addition it also contains bicarbonates. The liver secretes Bile which is stored in the gall bladder until released into the duodenum.

The Bile contains Bile salts which act as emulsifiers that help in fat digestion and absorption.

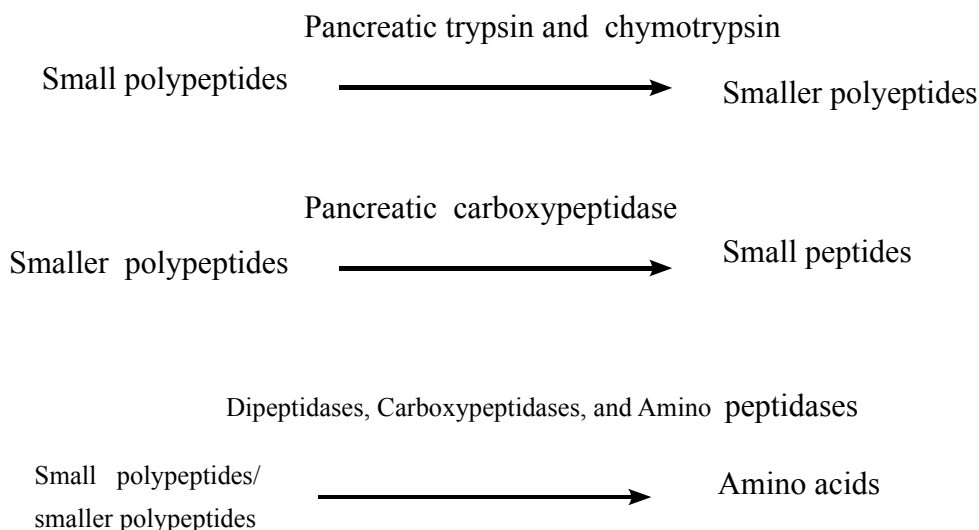
Carbohydrate digestion

Pancreatic amylase catalyze the conversion of polysaccharides (e.g. starch) into disaccharides. Intestinal disaccharidases catalyze the conversion of disaccharides into monosaccharides.



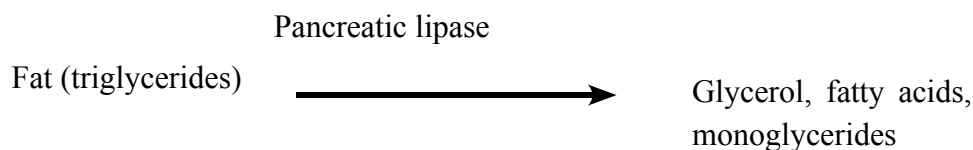
Protein digestion

Trypsin and Chymotrypsin catalyse the conversion of small polypeptides into smaller polypeptides. These smaller polypeptides are converted to small peptides and amino acids by the catalytic action of Pancreatic carboxypeptidases. Proteases secreted by the intestinal epithelium (Dipeptidases, Carboxypeptidases and Aminopeptidases) catalyse the conversion of small peptides into amino acids.



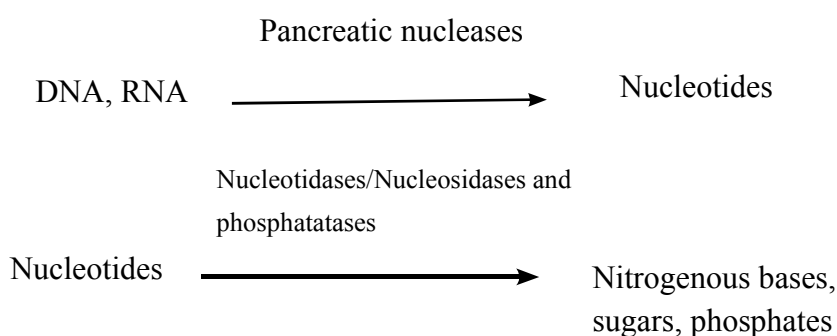
Fat digestion

Digestion of fats (triglycerides) starts in the small intestine. First bile salts emulsify fats. Next Pancreatic lipase catalyse the conversion of these fats into fatty acids, glycerol and monoglycerides.



Nucleic acid digestion

Digestion of nucleic acids start in the small intestine. Pancreatic nucleases catalyse the conversion of DNA and RNA into nucleotides. These nucleotides are eventually converted into nitrogenous bases, sugars and phosphates by the catalytic action of intestinal Nucleotidases, Nucleosidases and Phosphatases.



Absorption in the small intestine

- For effective absorption, the surface area of the intestinal wall has been increased with three structural modifications: heavy permanent foldings, finger like projections called villi in the intestinal wall and finger like microscopic projections called microvilli in the epithelial cells of the villi. These micro villi are exposed into the intestinal lumen, it gives the appearance of brush (brush border).
- Transport of the nutrients across the epithelium may be active or passive. For example Fructose is absorbed by facilitated diffusion. Amino acids, small peptides, vitamins, and most glucose molecules are actively transported into the epithelial cells.
- Then these nutrients from the epithelial cells are transported into the blood capillaries in the villi. Those blood capillaries are converged into the hepatic portal veins. These nutrients are carried in to the liver via the hepatic portal veins. From the liver, this nutrient filled blood is transported into the tissues.
- But absorption of some products of fat digestion takes place in a different pathway: Fatty acids and monoglycerides are absorbed into the cell through microvilli. Within the cells triglycerides are reformed and they are incorporated into water soluble globules called chylomicrons. These chylomicrons are transported into the lacteal and then into the blood vessels through lymph. Then they are transported throughout the body via the circulatory system.

- In addition to nutrient absorption, recovery of water and ions mostly occur in the small intestine. In addition to the water intake (about 2 L) digestive juices add more water (about 7 L) into the small intestine. Most of this water is reabsorbed via osmosis.

Large intestine – The large intestine is the terminal end of the alimentary canal. It is divided into three regions: colon (proximal part), cecum and rectum. The small intestine is connected to the large intestine at a ‘T’ shaped junction. One arm of the ‘T’ junction is colon and the other arm is a small pouch called cecum. A finger like projection in the cecum is called the appendix. The colon leads to the rectum and anus. Cecum is important for fermentation of indigested materials by microbes, especially in animals that eat large amount of plant matter.

Functions of the large intestine

- The colon: completes the reabsorption of water, synthesize some Vitamin B complexes, Vitamin K and folic acid with the help of microbes and move feces (consists of undigested matter such as fibres) along the colon by peristalsis.
- The rectum stores feces until they are eliminated. Presence of two sphincters between the rectum and anus can regulate feces movement. Strong contractions in the colon trigger the defecation.

Associated glands

Pancreas

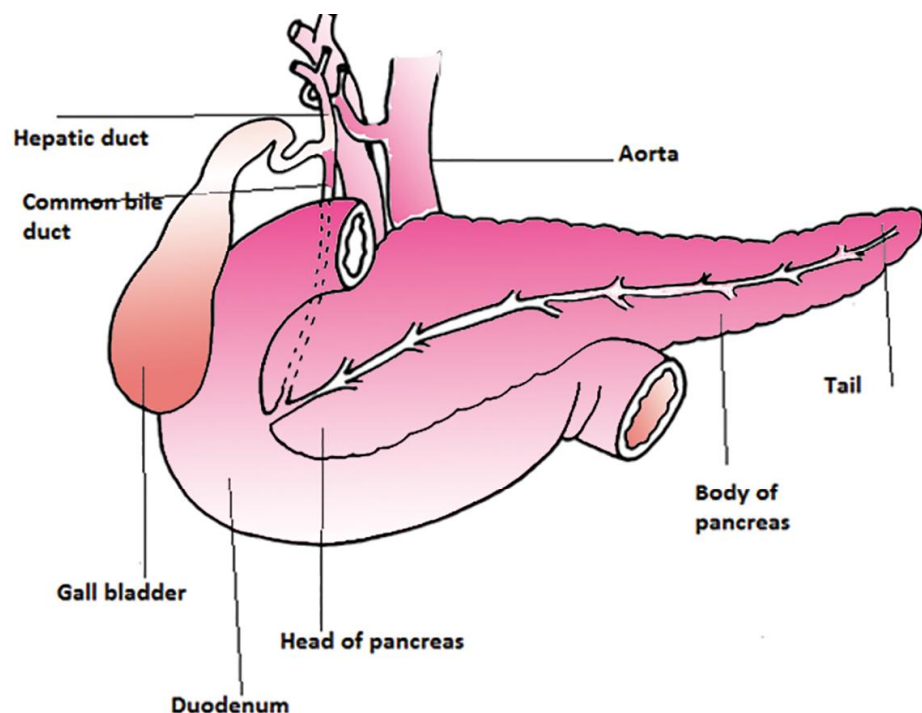


Fig 5.10: The pancreas in relation to the duodenum and biliary tract

Histological structure of pancreas

The pancreas is a pale grey gland which consists of a broad head, a body and a narrow tail. Head is in the curve of the duodenum. Pancreas is both an exocrine and endocrine gland.

The exocrine part consists of a large number of lobules made up of small acini, the walls of which consist of secretory cells. Each lobule is drained by a tiny duct and these unite eventually to form the pancreatic duct which joins with bile duct to form hepato pancreatic duct and opens into the duodenum at its midpoint. Exocrine part of the pancreas secretes pancreatic juice. The components of the pancreatic juice are bicarbonate, carbohydrate digesting enzymes (pancreatic amylase), pancreatic lipase, nucleases and inactive form of protein digesting enzymes (trypsinogen and chymotrypsinogen). These inactive enzymes are converted to active enzymes (trypsin and chymotrypsin) upon secretion into the lumen of the duodenum.

Endocrine part of the pancreas is the islets of Langerhans, which consist of group of specialized cells. They do not have ducts. Islets of Langerhans secrete hormones, glucagon and insulin which are involved in glucose homeostasis.

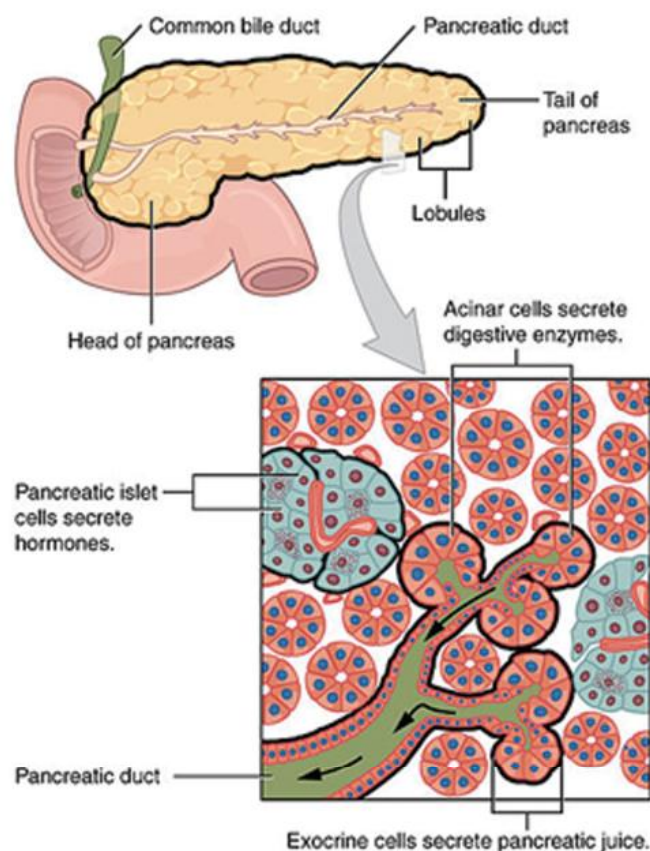


Fig 5.11: Histological structure of the pancreas

Liver

Liver is the largest gland in the body. Its upper and anterior surfaces are smooth and convex. Its posterior surface is irregular in outline. Liver contains four lobes. Each lobe is made up of tiny hexagonal shape lobules which are the functional unit. These lobules are made up of cuboidal cells called hepatocytes which are arranged in pairs of columns radiating from a central vein. Between two pairs of column of cells there are sinusoids (blood vessels with incomplete walls) containing mixture of blood from the tiny branches of the portal vein and hepatic artery. This arrangement allows venous blood (high concentration of nutritional materials) to mix with arterial blood and come into close contact with liver cells. Hepatic macrophages (Kupffer cells) are found in the lining of the sinusoids. Blood drains from the sinusoids into central veins which joins with veins from the other lobules, forming larger veins and eventually the hepatic vein. (fig 5.12). Bile canaliculi run between columns of liver cells. Canaliculi join up to form larger bile canals. In the corner of the hexagonal structure a branch of hepatic artery, a branch of the hepatic portal vein and intra lobular bile duct can be found.

Liver is a vital organ that performs many important functions. In addition to its role in food digestion it is also perform functions such as metabolism of carbohydrates, fats and proteins, detoxification of drugs and toxic substances, defense against microbes, some hormone inactivation and heat production.

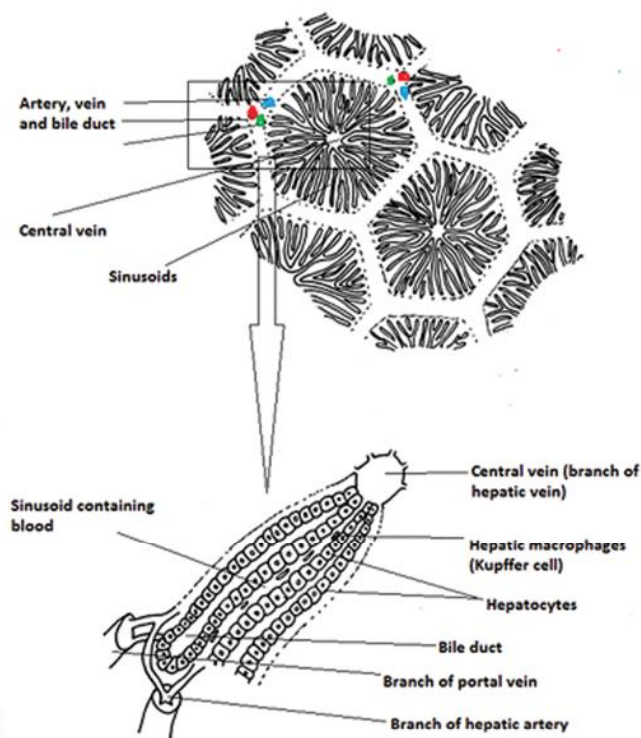


Fig 5.12: Histological structure of the liver

Function of liver related to digestion

The liver secretes bile which is stored in the gall bladder until released into the duodenum. The bile contains bile salts which act as emulsifiers that help in fat digestion and absorption.

Most of the absorbed nutrients reach the liver and it regulates the distribution of nutrients to the rest of the body. Excess glucose is stored as glycogen in the liver cells. Glycogen deposition and break down in the liver cells are regulated by insulin and glucagon hormones. Fat soluble Vitamins (A,D,E and K) and some water soluble vitamins (B12), iron and copper are also stored in liver.

Regulation of digestion in man

Digestion in man is regulated by two ways: nervous regulation and endocrine regulation. Nervous regulation is mainly by nervous reflexes. For example nervous reflex stimulates the release of saliva when food reach the mouth. Arrival of food in the stomach trigger churning and release of gastric juices. Endocrine system plays a critical role in digestion especially in the stomach and small intestine.

when food arrives the stomach, the stomach wall is stretched. This triggers to release the hormone gastrin. Gastrin circulates via the blood stream and arrives the stomach. Then gastrin stimulates the production of gastric juice at the stomach.

Fatty acids and amino acids in the chyme trigger the release of Cholecystokinin and Secretin from the duodenum. Cholecystokinin triggers release of bile from the gall bladder and digestive enzymes from the pancreas. Secretin stimulates the release of bicarbonate from the pancreas. Bicarbonate neutralize the chyme received from the stomach

When the chyme is rich in fat, food digestion in the stomach slows down due to high levels of Cholecystokinin and Secretin secreted by duodenum. These hormones act on the stomach and inhibit peristalsis and gastric juice secretion.

Balanced Diet

The balanced diet contains the all essential nutrients required for health in the appropriate proportions. Essential components of the balanced diet contain carbohydrates, proteins, lipids, fibers, minerals, vitamins and water. Carbohydrates and lipids supply energy. Energy requirements vary with age, sex, body size and activity. Twenty amino acids are needed to synthesize proteins in the body. Most of these amino acids are synthesized within the body. They are known as non essential amino acids (e.g. Alanine, cystine, etc). Other amino acids (e.g. lysine and histidine, etc.) must be obtained from the diet as they can't be synthesized within the body. They are called essential amino acids. Animal proteins contain all the essential amino acids in proper proportions. However most plant proteins lack one or more essential amino acids. Therefore vegetarian based diet requires several sources of plant proteins to obtain all the essential amino acids.

Components of foods and their functions

- Carbohydrates
- Proteins
- Lipids
- Vitamins
- Mineral elements
- water
- fibers

Carbohydrates

Carbohydrates are sugars and polysaccharides. They can be found in a variety of foods such as rice, bread, biscuits, cereals, hoppers, etc. during digestion most carbohydrates are broken down into monosaccharides which are absorbed into blood stream.

Functions of digestible carbohydrates

- Provide energy and heat: breakdown of carbohydrates provides ATP for body functions and generates heat.
- Act as an energy stores. e.g. excess carbohydrates are converted into glycogen and fat
- Facilitates protein sparing- proteins are not used to get energy when there is an adequate carbohydrates in the food

Proteins

Proteins are made up of amino acids during the digestion, proteins are broken down into amino acids and absorbed into the blood stream.

The amino acids are grouped into two groups i.e. essential amino acids and non essential amino acids. These essential amino acids cannot be synthesized in the body, therefore they should be obtained into the body through the diet. The non essential amino acids can be synthesized within the body. Therefore it is not necessary to obtain them through the diet.

Functions of proteins in the diet

- The amino acids which are supplied from proteins
 - are used for growth and repair of body cells and tissues
 - are used for synthesis of plasma proteins, enzymes, antibodies and some hormones
- Act as an energy source for body functions

Lipids

Lipids in the diet are mainly composed of fats and oils. Fatty acids are composed of fats and oils. Fatty acids can be grouped as essential and non essential fatty acids. Essential fatty acids cannot be synthesized within the body while the non essential fatty acids can be synthesized within the body. Therefore essential fatty acids should be obtained through the food.

Functions of lipids in the diet

- Provide energy and heat (on weight basis fats and oils provide more energy compared to carbohydrates and proteins)
- Help in transport and storage of fat soluble vitamins such as Vitamin A, D, E and K
- Store energy as fat in the adipose tissues
- Help to synthesize steroid hormones from cholesterol.
- Provide insulation: (e.g fat found in subcutaneous layer in the skin reduces heat loss , constituents of myelin sheath of neurons)

Vitamins

Vitamins are organic compounds required in small amounts for the maintenance of normal health and metabolism. Vitamins cannot be produced in the body and therefore should be provided in the diet. If the vitamins are insufficiently taken into the body, that may lead to the deficiency diseases. Vitamins are two types they are fat soluble vitamins (Vitamin A, D, E and K) and water soluble vitamins (Vitamin B and C).

Main Functions of Vitamins

- Vitamin A- form visual pigments in the eye, epithelial tissue maintenance, promotion of growth and immunity
- Vitamin B- components of coenzymes such as FAD and NAD, promote red blood cell production
- Vitamin C- act as an antioxidant, used in collagen synthesis
- Vitamin D- aids in absorption and use of Calcium and Phosphorous
- Vitamin E- act as an antioxidant
- Vitamin K- important in blood clotting

Minerals

Minerals are inorganic substances and they are also important for normal health and many body functions. Major mineral elements needed by humans are Ca, P, S, K, Cl, Na, Mg, Fe, F and I . In addition minerals needed in trace amounts include Co, Cu, Mn, Mo, Se and Zn.

Main functions of minerals include

- Ca- form bones and teeth, helps clotting blood and nerve and muscle function
- P- form bones and teeth, help maintain in acid base balance
- S- components of some amino acids
- K- help maintain in acid base balance and water balance, nerve function
- Cl- help maintain in acid base balance, maintain osmotic balance, nerve function
- Na- help maintain in acid base balance and water balance, nerve function
- Mg- act as enzyme cofactor
- Fe- components of hemoglobin and electron carriers, act as an enzyme cofactor
- F- maintenance of tooth structure
- I- component of thyroid hormone

Water

Water accounts for around 60% of the body mass in humans. Normally water is lost through urine, sweating and feces. Therefore amount of water in the body should be balanced within the body. This water can be taken in to the body by drinking and via foods.

Functions of water in human body

- Provides the moist internal environment for all living cells.
- Major component of blood and tissue fluid therefore helps to transport materials around the body and to exchange materials between blood and tissues and body cells
- Regulate body temperature mainly through evaporative cooling
- Dilute waste products and toxins and provide a medium for their excretion
- Moistens the food and that make easier to swallow

Fibres

Dietary fibres (non starch polysaccharides) are made up of indigestible polysaccharides in the diet. Fibres are rich in fruit, vegetable and cereals.

Functions of dietary fibres

- Provide bulk to the diet and satisfy the appetite.
- Prevent constipation by attracting water to increase faecal bulk and stimulating peristalsis leading to defecation.
- Adequate fibres in the diet protect against some gastro-intestinal disorders such as cancers in the colon and rectum.

Essential nutrients: Essential nutrients are the substances that cannot be synthesized in the body from simple precursors and must therefore be taken through the diet. These essential nutrients include essential amino acids, essential fatty acids, vitamins and minerals. Essential nutrients have key functions in bio synthetic reaction in the body cells. If these essential nutrients are not supplied in correct proportions in the diet that will lead to malnutrition. Therefore it is essential to obtain them in correct amounts.

Essential amino acids: Essential amino acids are the amino acids that must be obtained from the food since they cannot be synthesized within the body from organic precursor molecules. Of the 20 amino acids required to make proteins in the body 8 amino acids are essential amino acids. Examples for essential amino acids are leucine and methionine. The animal protein products (e.g. eggs, meat, cheese, etc.) will provide all essential amino acids in correct proportions required for body functions. Most plant proteins are “incomplete” as they are deficient in one or more amino acids. Therefore vegetarian diet should contain a variety of plant proteins in order to obtain all the essential amino acids required.

Essential fatty acids: Essential fatty acids are the fatty acids that should be obtained from the diet since they cannot be synthesized in the body from organic precursors. Seeds, grains and vegetables provide enough amounts of essential fatty acids.

Table : Dietary sources and deficiency symptoms of vitamins and minerals

Vitamin/ Mineral	Main dietary sources	Deficiency symptoms
Fat soluble vitamins		
Vitamin A (retinol)	Dark green vegetables, orange vegetables and fruits, dairy products	Blindness, skin disorders, immunity impairment
Vitamin D	Egg yolk, dairy products	Bone deformities(rickets) in children, bone softening in adults
Vitamin E	Vegetable oils, nuts, seeds	Nervous system degeneration
Vitamin K	Green vegetables, tea, produced by colon bacteria	Defective blood clotting
Water soluble vitamins		
Thiamine (Vitamin B ₁)	Legumes, peanuts, whole grains, pork	Beriberi(characterized by tingling, poor coordination, susceptibility to infection, reduced heart function)
Riboflavin (Vitamin B ₂)	Dairy products, meats, vegetables, enriched grains	Skin lesions (cracks at corners of mouth)
Niacin (Vitamin B ₃)	Grains, nuts, meats.	Pellagra(characterized by lesions in skin, mental confusion and diarrhea)
Pantothenic acid (Vitamin B ₅)	Dairy products, fruits, vegetables, grains	Fatigue, numbness, tingling of hands and feet

Vitamin/ Mineral	Main dietary sources	Deficiency symptoms
Pyridoxine (Vitamin B ₆)	Whole grains, Meats, vegetables	Irritability, anemia
Biotin (Vitamin B ₇)	Meats, legumes, vegetables	Neuro- muscular disorders, scaly skin inflammation
Folic acid (Vitamin B ₉)	Green vegetables, whole grains	Anemia, birth defects
Cobalamin(Vitamin B ₁₂)	Dairy products, eggs, meats	Loss of balance, numbness, anemia
Ascorbic acid (Vitamin C)	Citrus fruits, broccoli, tomatoes	Scurvy (characterized by degeneration of skin and teeth) , delayed wound healing
Minerals		
Calcium (Ca)	Dairy products, dark green vegetables, legumes	Loss of bone mass, impaired growth
Iron (Fe)	Whole grains, green leafy vegetables, legumes, meats, eggs	Anemia, weakness, impaired immunity
Phosphorus (P)	Rice, bread, milk, dairy products, fish, red meat	Decaying of teeth and bones, weakness
Potassium (K)	Fruits, vegetables, meat , dairy products, grains	Muscle weakness, nausea , paralysis, heart failure
Iodine(I)	Sea foods , vegetables, iodized salt	Goiter(enlarged thyroid glands)
Sulfur (S)	Foods containing proteins	Fatigue, Impaired growth, swelling
Chlorine (Cl) and Sodium (Na)	Table salt	Reduced appetite, muscle cramps
Magnesium (Mg)	Green leafy vegetables, grains	Disturbance in nervous system
Fluorine (F)	Tea, sea food, drinking water	Tooth decay

Basal metabolic rate (BMR)

- Basal metabolic rate is defined as the minimum metabolic rate at rest, when in a post absorptive stage (at least 12 hour fasting) and is not experiencing stress.
- BMR is measured under a “comfortable” temperature range.
- BMR of humans averages for adult males 1,600-1,800 kcal per day and 1,300-1,500 kcal for adult females.

Energy budget: An energy budget is a balance sheet of energy intake against energy expenditure in a particular animal. The basic model of energy budget can be shown as follows;

$$C=M+U+F+P$$

Where C=Energy content in the food sources taken in

M= Energy spent for metabolic activities

U= Energy associated with urinary loss

F= Energy associated with fecal loss

P= Production (Energy available for growth and reproduction)

In energy budgets, energy content in the food intake is compared with energy expenditure which includes energy spent for basal metabolism and extra activities (M), energy associated with excretory products namely urinary loss (U) and fecal loss (F). The energy differences between the energy intake and the energy expenditure for metabolism and excretion are available for production which includes growth and reproduction. Energy budget can be calculated for each animal based on energy measurements from field and laboratory. Energy budgets are useful for estimating energy available for growth and reproduction.

Food for healthy life: For a healthy life diet should contain correct proportions of carbohydrates, proteins, lipids, water, fiber, essential mineral elements and vitamins. Dietary deficiencies can have negative impact on health. When food intake exceeds daily energy requirements especially in inactive individuals can lead to ill health conditions specially diabetes mellitus and heart diseases. Some individuals develop allergic reactions to foods such as pineapple, peanuts and tomatoes. Such individuals should avoid these types of foods. Antioxidants present in the food material (e.g. vitamin C and vitamin E) are also important in avoiding disorders in the alimentary canal and maintaining a healthy life. As humans cannot synthesize all the required antioxidants some of them should be obtained from the diet.

Malnutrition: Malnutrition can arise due to failure of obtaining an adequate nutrition when the diet lacks one or more essential nutrients or consistently supplies less chemical energy than the energy required by the body. According to the WHO if, body mass index (BMI) is less than 18.5 is said to be malnutrition. BMI of a person is calculated as follows,

$$\text{BMI} = \text{Mass} / \text{height}^2 \text{ (kg/ m}^2\text{)}$$

Obesity: Obesity arises when energy expenditure of a person is much less than the energy intake. According to the WHO criteria, if the BMI is at 30.0 or over it is known as obesity. This condition is a growing issue worldwide. Obesity can lead to many diseases such as diabetes mellitus, cardiovascular diseases, some cancers etc.

Common disorders in the alimentary canal

Gastritis: gastritis is a condition resulting inflammation of the stomach which can be due to several reasons. In gastritis, glands of the stomach wall are stimulated and secrete excess HCl causing damage to the mucosa. Due to the damages of mucosa layer of the stomach, blisters can be formed. Prolonged starvation and mental stress are one of the reasons for the secretion of excess HCl. Some drugs like aspirin can also induce gastritis conditions. Longer lasting gastritis conditions is usually associated with the infection by the acid tolerance bacterium *Helicobacter pylori*. As prolonged starvation is one of the reasons for developing gastritis proper food habits should be practiced to control this condition.

Constipation: Constipation occurs due to the slow movement of feces that promotes the water reabsorption and as a result feces become more solid. Inhibition of the reflex action in defecation may also lead to constipation. This leads to pain in the anus and difficulty in defecating. Constipation can be controlled by developing behavioral adjustments to carry out defecation properly. Intake of adequate fiber in the diet can help prevent constipation.

Circulatory Systems in Animals

Need of a circulatory system

A circulatory system is required in animals for transportation of materials within the body and exchange of materials with the external environment. The simple animals (e.g. Cnidarians, Flat worms) lack specialized system for the transport and distribution of material because many or all cells are in direct contact with the external environment. In these animals exchange of materials over the body surface through direct diffusion is sufficient for their needs. In these animals those materials can be transported by diffusion through short distance within the body.

As organisms increase in size and complexity, the amount of materials moving in and out of the body also increases. The distance that materials have to be transported within the body also increases and many cells are not in direct contact with their external environment. Therefore diffusion is not sufficient to transport materials throughout the body. Therefore circulatory systems have evolved in such organisms to exchange materials between cells and their immediate surroundings.

Substances that are transported in the body

Substances that are transported in the body include respiratory gases (oxygen, carbon dioxide), nutrients (glucose, amino acids, fatty acids, vitamins, etc.), waste products of metabolism (urea, ammonia, etc), hormones and antibodies.

Blood circulatory systems in the animal kingdom

A circulatory system has three basic components: a muscular pumping device (heart), interconnected vessels and a circulatory fluid (blood/ hemolymph). Circulatory fluid flows through the vessels mainly due to the pressure generated by the heart. The circulatory system functionally connects the fluid environment of the body cells to the organs that exchange gases, absorb nutrients and dispose of wastes by transporting fluids throughout the body. Various degrees of complexity in the arrangement of circulatory systems can be seen in the animal kingdom.

Main circulatory systems in the animal kingdom

There are two types of circulatory systems among the animals: open circulatory system and closed circulatory system.

The open circulatory system: It is a circulatory system in which a fluid called hemolymph that bathes the tissues and organs directly. There is no distinction between the circulatory fluid and the interstitial fluid surrounding cells. The heart pumps hemolymph through the circulatory vessels into the spaces (interconnected sinuses) which surround body tissues. Chemical exchange occurs directly between the hemolymph and body cells. Back flow of the hemolymph takes place via the pores (ostia) with valves found in the heart during the relaxation. The open circulatory system has evolved in some invertebrate phyla such as Arthropoda and Mollusca (some mollusc groups).

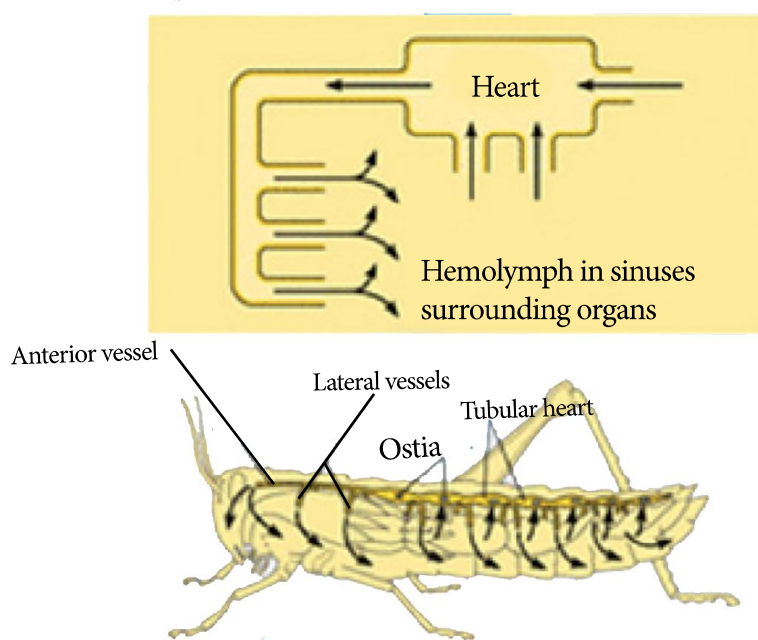


Fig 5.13: Circulatory system of a Grass hopper

The closed circulatory system: It is a circulatory system in which blood is restricted to vessels and kept apart from the interstitial fluid. Blood is pumped by the heart/s into large vessels. These large blood vessels branches into small vessels and they penetrate into the organs. Chemical exchange occurs between the blood and the interstitial fluid and interstitial fluid and body cells. It may contain one or more hearts. This type of circulatory systems can be seen in vertebrates and invertebrates such as Annelids. In comparison to open circulatory system enable effective delivery of oxygen and nutrients to the cells of larger and more active animals due to relatively high blood pressure.

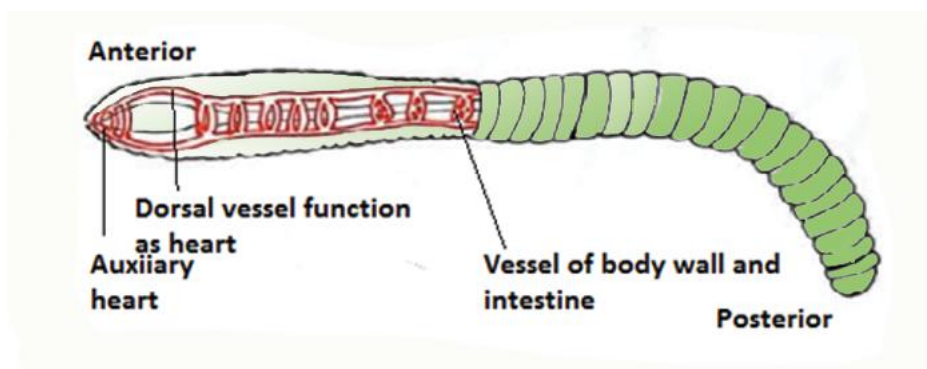


Fig 5.14: Closed blood Circulatory system of an Annelid

Organization of vertebrate circulatory systems: Single circulation and double circulation

Vertebrates have a closed circulatory system where there are three main types of blood vessels: arteries, veins and capillaries. Within each type blood flows in only one direction. The blood vessels which carry blood from the heart to organs are called arteries. When these arteries branch into smaller vessels within organs they are called arterioles. They pass blood to capillaries which are microscopic vessels with thin and porous walls. These are the places where materials exchange occurs between blood and interstitial fluid surrounding body cells through diffusion. Capillaries converge into venules and they converge into veins which carry blood back to the heart.

Single circulation

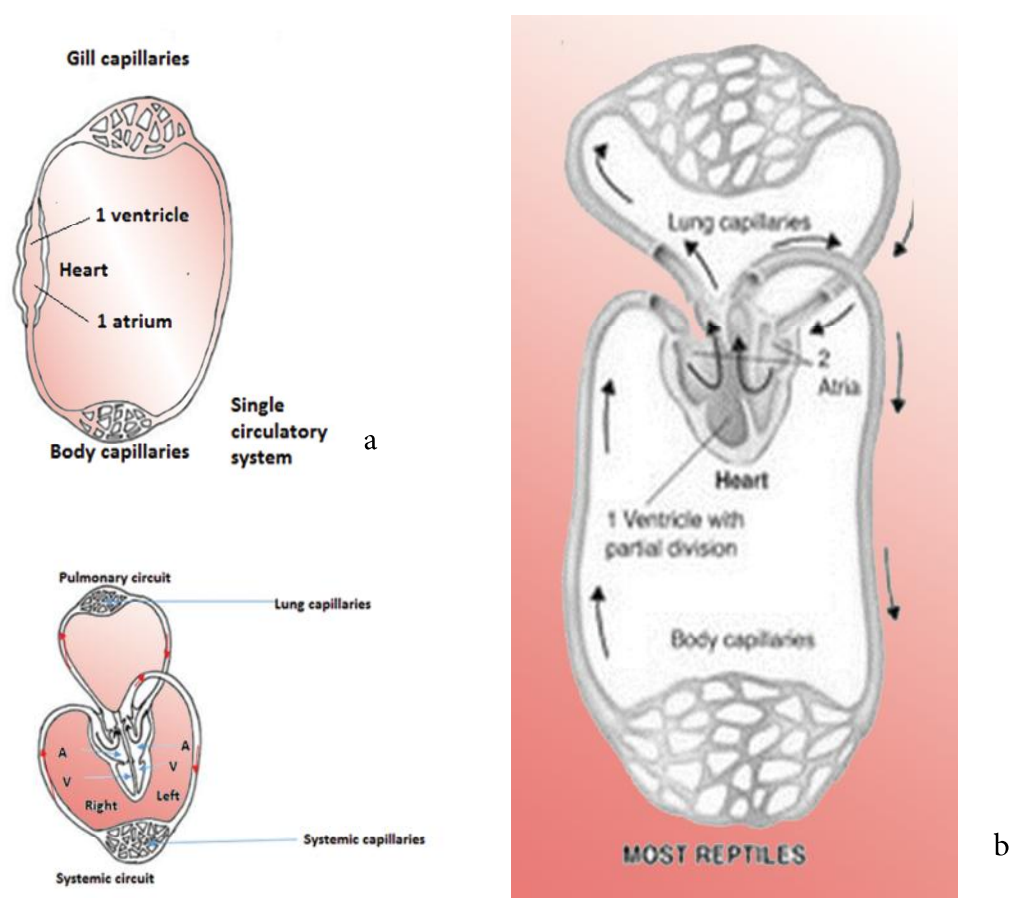
During single circulation, in a complete circulation through the entire body, blood passes through the heart only once. In animals that demonstrate single circulation, possess two chambers in the heart. They are atrium and ventricle.

During a single circulation, the oxygen poor blood returning from the body enters atrium and pass into the ventricle. Then the ventricle is contracted, blood is pumped into the capillary bed in the gills. There gas exchange occurs between the capillaries and the external environment. The O_2 is diffused into the blood while CO_2 is removed from the blood. Next the oxygen enriched blood circulates throughout the body and reaches the body cells through body capillaries. e.g.: Bony fishes, Cartilaginous fishes such as rays and sharks.

Double circulation

During double circulation, in a complete circulation through the entire body, blood passes through the heart twice. Such circulatory system consists of separate pulmonary and systemic circuits in which blood passes through the heart after completing each cycle. e.g. Amphibians, Reptiles, Aves, Mammals. Amphibians and most reptiles have a three chambered heart: two atria and one ventricle. Birds and mammals have a four chambered heart where heart is completely divided into left and right sides. This arrangement allows the complete separation of oxygen enriched and oxygen poor blood. The oxygen poor blood from systemic circulation flows to the right atrium of the heart and then to the right ventricle. Right ventricle pumps blood into the lungs. Oxygen enriched blood from the lungs reaches the left atrium and pass to the left ventricle. The left ventricle pumps oxygen enriched blood into the systemic circulation.

Double circulation is more effective in supplying blood to all body organs and tissues especially brain and muscles due to the higher pressure exerted by the heart in the systemic circulation. This in contrast to single circulation where blood flows under reduced pressure from the gas exchange organs to other organs.



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Fig 5.15: Single and double circulatory systems in animals: a. Single circulation (fish), b. double circulation (amphibian) c. double circulation (mammal)

Basic plan of human blood circulatory system and lymphatic system

Basic plan of human blood circulatory system

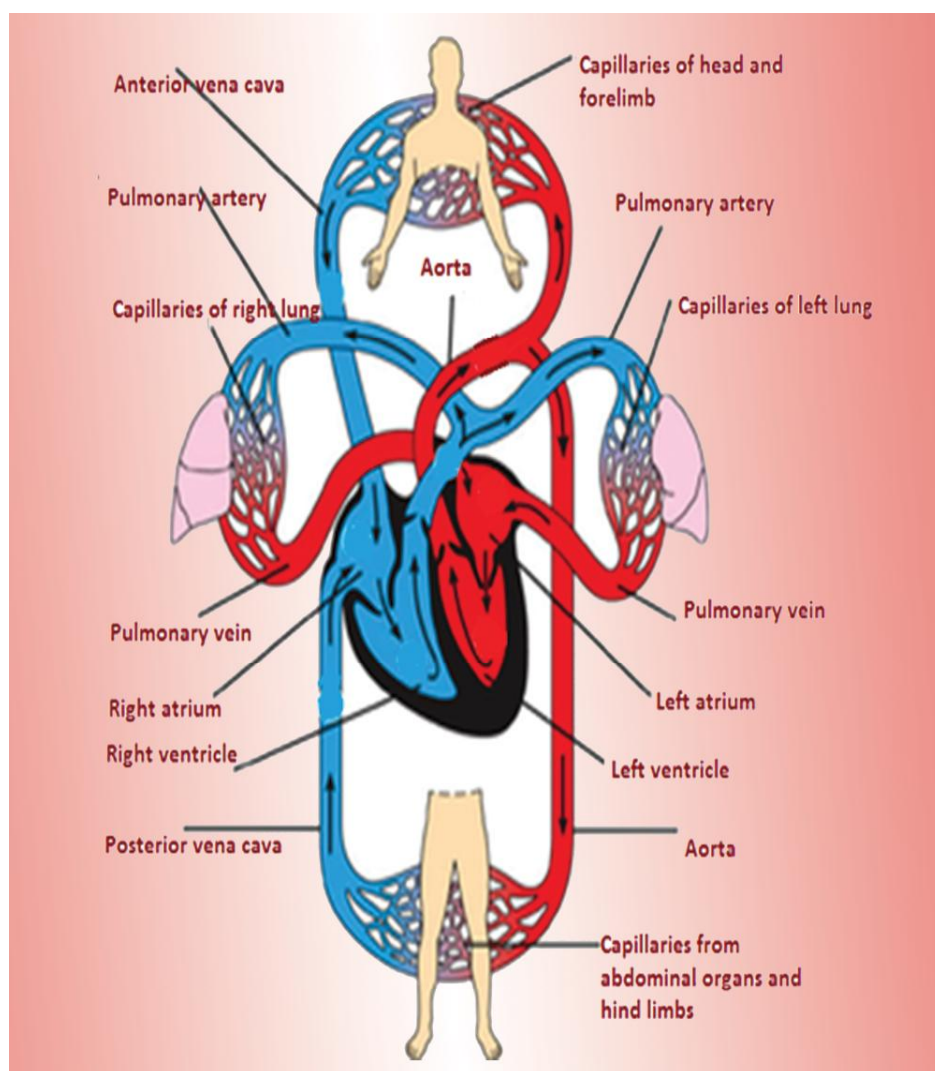


Fig 5.16: Basic plan of the human blood circulatory system

Figure (5.16) depicts the basic plan of the human blood circulatory system. The human heart consists of four chambers; they are two ventricles and two atria. There is a dual circuit which operates simultaneously. The pulmonary circuit takes oxygen poor blood to the respiratory surface, the lungs and returns the oxygen enriched blood back to the heart while the systemic circuit supplies oxygen enriched blood to all organs and tissues in the body and returns oxygen poor blood from organ and tissues back to the heart. Each circuit consists of major arteries/arteries, arterioles, capillary beds, venules and veins/major veins.

During ventricular contraction, the right ventricle pumps the oxygen poor blood into the two lungs via the pulmonary arteries. Then in the lungs O_2 is loaded into the blood through diffusion while CO_2 is unloaded from blood into the external environment.

This process occurs in the capillary beds in the left and right lungs. Then the oxygen rich blood is transported into the left atrium via the two pulmonary veins. During atrial contraction, this oxygen rich blood is transported into the left ventricle. During ventricular contraction oxygen rich blood is pumped into the aorta. Through arteries the aorta conveys this oxygen rich blood throughout the body. First the aorta branches into the coronary arteries which supplies blood into the heart muscles. Then the aorta branches into arteries, arterioles leading to capillary beds in the head and arms and the capillary beds in the abdominal organs and legs. The exchange of gases occurs in the capillary beds where O_2 rich blood is diffused into the tissues while the CO_2 rich blood diffused into the blood capillaries. These blood capillaries rejoined to form venules which direct oxygen poor blood into veins. Oxygen poor blood from trunk and hind limbs is drained into the inferior vena cava and oxygen poor blood from the head, neck and fore limbs are directed into the superior vena cava. The blood from both inferior and superior vena cava is pumped into the right atrium where it is passed into the right ventricle. This blood is directed to the pulmonary circuit as explained above.

Basic plan of human lymphatic system

The lymphatic system is closely connected with the blood circulatory system both structurally and functionally. It consists of lymph vessels through which lymph travels. Other structure in the lymphatic system includes lymph nodes, lymphoid tissues (tonsils) and lymphoid organs (e.g. spleen and thymus). Lymph vessels consist of tiny vessels and larger vessels. Tiny lymph vessels are in close contact with the capillaries of the blood circulatory system. Lymph nodes are composed of connective tissues and white blood cells.

The lymphatic system returns lost fluid and proteins from the blood capillaries back into the blood. The lost fluid from the blood capillaries is called lymph when they are inside the lymphatic system. The composition of the lymph is same as interstitial fluid. Lymph vessels have valves. That prevents the backflow of the lymph. The lymph is drained into veins at the base of the neck via two large ducts. The rhythmic contraction of the lymph vessel walls and skeletal muscle contraction help to move the lymph.

The functions of human lymphatic system include tissue drainage to maintain the blood volume in the blood circulatory system, absorption of fat and fat soluble vitamins from the small intestine and for immune responses.

Structure and function of the human heart

The human heart is roughly a cone shaped hollow muscular organ. The heart wall is composed of three layers of tissues: Pericardium, Myocardium, Endocardium.

Pericardium: The pericardium is the outer most layers. It is made up of two sacs: The outer fibrous pericardium and inner serous pericardium.

Myocardium: Myocardium is the middle layer of the heart wall. It is composed of specialized cardiac muscle found only in the heart. Running through the myocardium is also a network of specialized conducting fibres responsible for transmitting the heart's electrical signals.

Endocardium: Endocardium is the inner layer of the heart wall. It lines the chambers and valves of the heart. It is a smooth membrane and consists of flattened epithelial cells. It is continuous with the endothelium lining of the blood vessels.

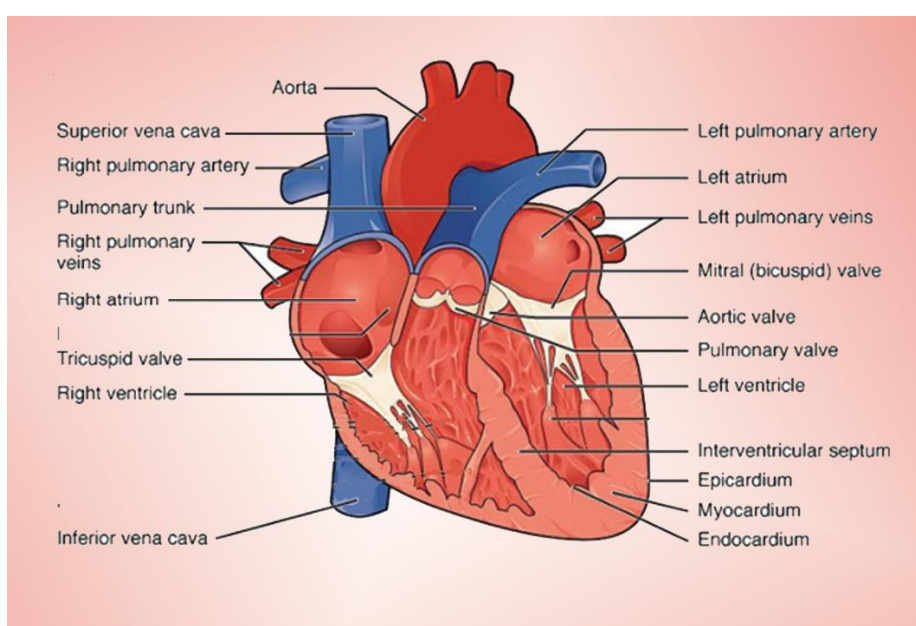


Fig 5.17: Interior of the human heart

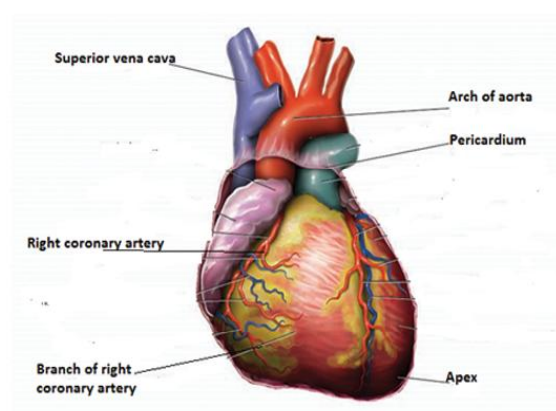


Fig 5.18: External appearance of the human heart

There are four chambers in the heart two upper atria and two lower ventricles. Ventricles have to pump blood to the whole body where as atria pump blood to the ventricles only. Therefore walls of the ventricles are thicker than walls of the atria. Wall of the left ventricle is thicker than walls of the right ventricle because the right ventricle pumps blood to the lungs which are closer to the heart whereas the left ventricle has to pump blood throughout the body.

Therefore the blood entering the aorta from the left ventricle is at much higher blood pressure than the blood entering the pulmonary artery from the right ventricle.

The heart is completely divided into a right and left side by a septum. The atria and ventricles on each side are divided by an atrio-ventricular valve (AV). The right atrioventricular valve has three flaps hence known as tricuspid valve and the left atrioventricular valve has two flaps hence known as bicuspid valve. Conical shaped papillary muscles are extensions of the inner wall of the ventricles. Atrio-ventricular valves attached to the papillary muscles by fibrous cords which are called chordae tendineae. Atrio-ventricular valves are anchored by strong fibres. This prevents the valves from being turned inside out. Semilunar valves are found at the points where the pulmonary artery and aorta leave the right and left ventricles respectively. These valves prevent the backflow of blood into the ventricles.

Two pulmonary arteries with oxygen poor blood leave the heart from the upper part of the right ventricle. Two pulmonary veins from each lung carry oxygen rich blood back to the left atrium. The aorta with oxygen rich blood leaves from the upper part of the left ventricles. The superior vena cava and inferior vena cava open into the right atrium and empty their contents into the right atrium. The heart is supplied with arterial blood by the right and left coronary arteries which branch from the aorta immediately after the aortic valve.

The conducting system of the heart

Heart generates its own electrical impulses and beats independently of nervous or hormonal control. However it is supplied with both sympathetic and parasympathetic nerve fibers which increase and decrease respectively the intrinsic heart rate. In addition to that heart responds to a number of circulating hormones including adrenaline and thyroxine.

Small group of specialized neuromuscular cells in the myocardium initiate and conduct impulses. The conducting system of the heart consists of following specialized system.

- SA node (Sinoatrial node)
- AV node (Atrioventricular node)
- Atrioventricular bundle (bundle of His), bundle branches and Purkinje fibres

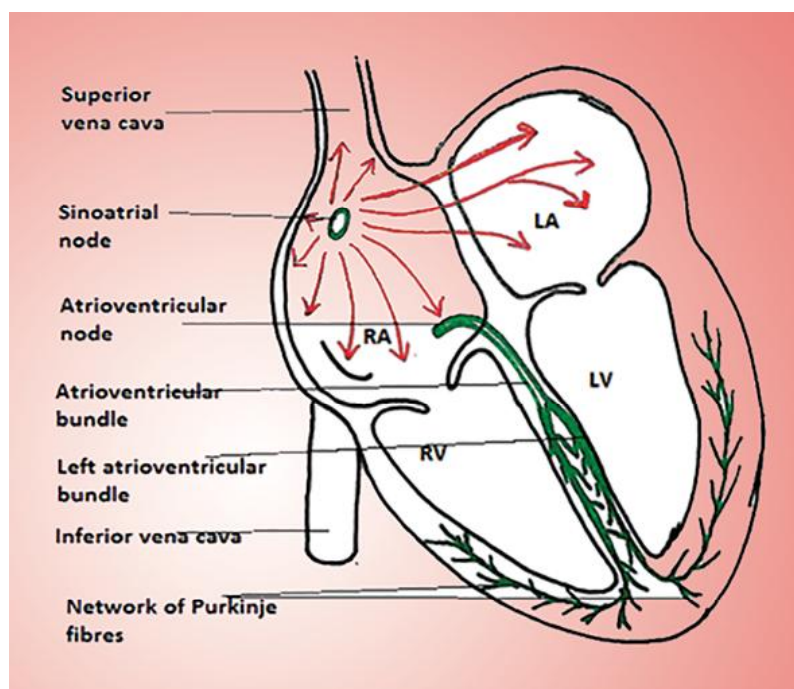


Fig5.19: The conducting system of the human heart

SA node / Sinoatrial node

SA node is a small mass of specialized cells. It lies in the myocardium of the right atrium near the opening of the superior vena cava. The stimulus for contraction of the heart originates in the 'SA node'. The SA node initiates the heart beat and sets the rhythm of the heart beat so it is called the pace maker. But the heart rate can be varied by the stimulation from the autonomic nervous system, hormones such as adrenaline, thyroxine and temperature.

AV node

AV node is also a small mass of specialized cells. It is situated between wall of the left and right atria. The AV node transmits the electrical signals from the atria into the ventricles.

Atrioventricular bundle (bundle of His), bundle branches and Purkinje fibres

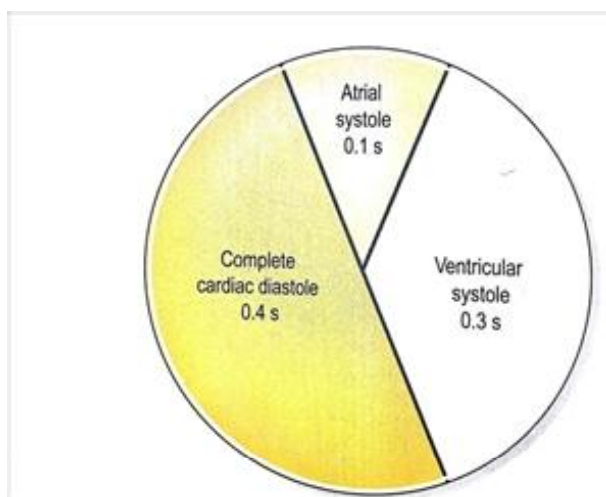
AV bundle is a mass of fibers. It originates from the AV node. The AV bundle crosses the fibrous rings that separate atria and ventricle at the upper end of the ventricular septum it divides into right and left bundle branches. Within the ventricular myocardium the branches break up into fine fibres. These fine fibers are Purkinje fibers. The AV bundle branches and the Purkinje fibers transmit electrical impulse from the AV node to the apex of the myocardium. As a result of this impulse, wave of ventricular contractions begin. Then the contraction sweeps upwards and outwards pumping blood simultaneously into the pulmonary artery and the aorta.

The cardiac cycle

The cardiac cycle refers to the sequences of events that take place in a complete heart beat. During this process one complete cycle of pumping and filling of blood into the heart occurs. Complete cardiac cycle lasts for 0.8 second. It occurs as follows:

1. Atrial systole – Contraction of the atria
2. Ventricular systole - Contraction of the ventricles
3. Complete cardiac diastole – Relaxation of the atria and ventricles

At rest the healthy adult heart is likely to beat at a rate of 60- 80 beats per minute. During a single heart beat, the heart contracts (systole) and then relaxes (diastole). During a single contraction the amount of blood pumped by a ventricle is called the stroke volume.



Total period of 1 cycle = 0.8 second

Fig 5.20: The stages of Cardiac cycle

Complete cardiac diastole

This lasts for 0.4 second. Both atria and ventricles are relaxed and blood return to the heart. The superior vena cava and the inferior vena cava transport oxygen poor blood into the right atrium. At the same time as the four pulmonary veins bring oxygen rich blood into the left atrium. The pressure of the atria is more than the pressure of the ventricles. Therefore atrioventricular valves are open and some blood flows passively through to the ventricles.

Atrial systole

When blood flows into the atrium SA node is stimulated. Then SA node triggers a wave of contractions that spreads over the myocardium of both atria. Hence the remaining blood in the atria flows into the ventricles thereby emptying the atria. This lasts for 0.1 second.

Ventricular systole

Through the atrial muscles the electrical impulses reaches the AV node. Then AV node triggers its own electrical impulses which quickly spread to the ventricular muscles via the AV bundle, the bundle branches, and Purkinje fibers. This results in a wave of contractions which sweep upwards from the apex of the heart across the wall of the ventricles. As a result both ventricles contract.

The pressure in the right ventricle is more than the pressure in the pulmonary artery and the pressure in the left ventricle is more than pressure in the aorta. Therefore pulmonary valve and aortic valves open and blood flows into pulmonary artery and aorta respectively.

The high pressure generated during ventricular contractions force the atrioventricular valves to close preventing backflow of blood into the atria. Ventricular systole lasts for 0.3 seconds. When ventricles relax, the pressure within them falls. The pulmonary and aortic valves close. Pressure within the pulmonary artery and aorta is more than pressure within the ventricles. The valves of the heart and great vessels open and close according to the pressure within the chambers of the heart. The sequence of opening and closing of valves ensure that blood flows only in one direction.

Electrocardiogram (ECG)

Electrical activity in the heart can be detected on the surface of the skin by placing electrodes on the chest or limbs as the body tissues and fluids conduct electricity well. Such a recording is called an electrocardiogram (ECG). The ECG indicates the spread of the electrical signal generated by the SA node as it travels throughout the heart. The normal ECG tracing of a healthy individual shows five waves which by convention have been named P, Q, R, S and T.

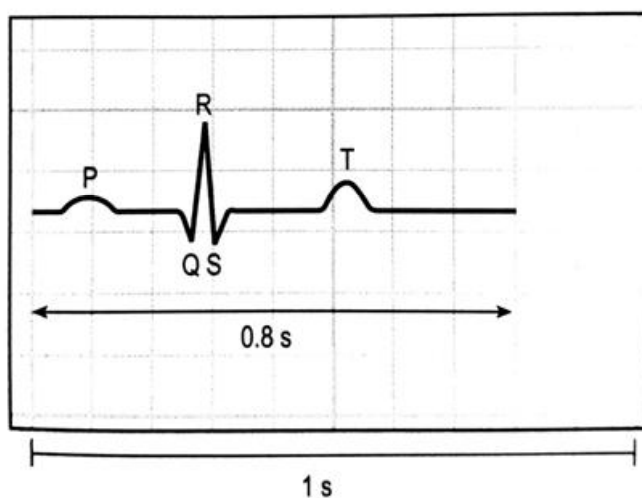


Fig 5.21: Electrocardiogram of one cardiac cycle

P wave - It represents the impulse from the SA node as it sweeps over the atria (atrial depolarization).

QRS wave complex – Represents rapid spread of impulse from the AV node throughout the ventricles and electrical activity of the ventricular muscles.(ventricular depolarization)

T wave – Represents ventricular repolarization and relaxation of the ventricular muscles. Due to the larger QRS complex, atrial repolarization which occurs during ventricular contraction is not seen.

Information about the heart function of a person (state of the myocardium and the cardiac conduction system) can be obtained by examining the pattern of waves and the time interval between cycles and parts of cycles.

Blood Pressure

The force that the blood exerts on the walls of blood vessels as it travels is referred to as blood pressure. Blood pressure in the arteries of systemic circulation maintains the essential flow of blood into and out of the organs of the body.

It is very important to keep blood pressure within normal limits. High blood pressure could lead to damage blood vessels resulting in formation of clots or bleeding from damaged sites. If the blood pressure falls too low, there will be inadequate blood flow through tissue capillary beds. This will adversely affects the normal functioning of vital organs such as the brain, heart and kidneys.

Blood pressure varies according to the time of day, the posture, gender, age, activity, exercise and stress (Emotional states) of an individual. Blood pressure falls at rest and during sleep. Blood pressure increases during excitement, fear or anxiety.

Systolic and diastolic pressure

Systolic pressure

Systolic pressure is the pressure produced within the arterial system when the left ventricle contracts and pushes blood into the aorta. At rest , systolic pressure in a normal healthy adult systolic pressure is about 120 mmHg.

Diastolic pressure

Diastolic blood pressure is the blood pressure within the arteries following ejection of blood at complete cardiac diastole(when the heart is at rest. In a normal healthy adult diastolic pressure is about 80 mmHg.

Arterial blood pressure is measured by a sphygmomanometer. It is expressed as

$$\frac{\text{Systolic pressure (mm Hg)}}{\text{Diastolic pressure (mm Hg)}} = 120/80 \text{ mmHg}$$

Hypertension and Hypotension

Hypertension

Sustained elevated blood pressure above normal limits is called hypertension.

Consequences of hypertension are kidney damage, adrenal gland disorders, heart attack (because of the increased heart rate and cardiac contraction), stroke (caused by cerebral haemorrhage), damaged blood vessels which can lead to death

Risk Factors for Hypertension

- Obesity
- Diabetes mellitus
- Family history
- Smoking
- A sedentary life style
- High intake of salts
- High intake of alcohol
- Stress
- Deposition of low density lipoprotein (LDL) on artery walls.

Hypotension

Sustained reduction of blood pressure below normal limits is called hypotension. Hypotension usually occurs as a complication of other condition such as shock, Dengue hemorrhage fever, standing up suddenly from sitting or lying position, over bleeding/ hemorrhage condition, fasting, low nutrition etc. Low blood pressure leads to inadequate blood supply to the brain. Depending on the cause unconsciousness may be brief (fainting) or prolonged possibly causing death.

Coronary circulation

The heart is supplied with arterial blood by the right and left coronary arteries which branch from the aorta immediately distal to the aortic valve.

The coronary arteries travel the heart wall eventually forming a vast network of capillaries. Most of the venous blood is collected into a number of cardiac veins that joins to form the coronary sinus which opens into the right atrium. The remainder passes directly into the heart chambers through little venous channels.

Consequences of blockage of coronary arteries

The inner lining of the arteries can be thickened and harden leading to the condition called atherosclerosis which occurs as a result of fatty deposits especially cholesterol particles. This can affect the normal blood supply to the organs and tissues.

One or more branches of coronary arteries can be blocked due to atherosclerosis which can be complicated by thrombosis (blood clot). Depending on the place (s) of the block in the coronary arteries and the degree of blockage related parts of the heart muscle will be deprived of oxygen and nutrients. Hence, narrowing of the arteries leads to chest pain (Angina). The complete occlusion due to the blockage of one or more coronary arteries leads to heart attacks (Myocardial infarction) which refers to the damage or death of cardiac muscle tissue due to lack of adequate oxygen and nutrients. Due to this, heart beat rhythm may be abnormal and the heart may cease to be an effective pump. The other vital organs such as brain may be deprived of inadequate supply of oxygen rich blood and heart attack may be fatal if not treated on time.

Stroke – Similarly blockage due to atherosclerosis or rupture of arteries supplying blood to the brain may cause the death of nervous tissue due to lack of oxygen and nutrients. This is referred to as stroke.

Respiratory pigments

Respiratory pigments are organic compounds which can combine with oxygen where the partial pressure of oxygen is high and release oxygen where partial pressure of oxygen is low. Since oxygen is less soluble in watery medium including blood, transportation of oxygen from respiratory surface to the tissues/ organs is a problem for complex animals. To overcome this problem animals have evolved the respiratory pigments.

Different respiratory pigments can be seen in the animal kingdom:

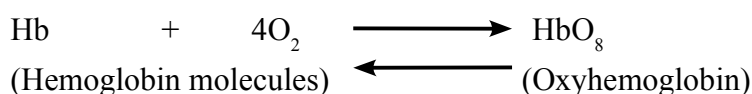
- Haemoglobin- present in blood of human, other vertebrate and annelids
- Haemocyanin- present in hemolymph of arthropods and molluscs.
- Chlorocruorin- present in the blood of many annelids
- Haemoerythrin- present in the blood of marine invertebrates (some annelids)
- Myoglobin- present in vertebrate muscles

All these respiratory pigments except myoglobin transport oxygen from respiratory surfaces to the tissues and organs and while transporting carbon dioxide from tissue/ organs to the respiratory surface for elimination. Myoglobin present in the muscle tissue has an oxygen storage function.

Transport of respiratory gases in human blood

Transport of oxygen

It is the hemoglobin molecule found in the erythrocytes which is responsible for the transport of oxygen around the body. Hemoglobin is composed of four subunits. Each subunit is composed of a globin protein and the haem group. Haem groups are responsible for the characteristic red colour of the blood. A ferrous (iron) atom is located within each haem group and each of these can combine reversibly with one molecule of oxygen. Therefore each hemoglobin molecule can carry up to four oxygen molecules.



Combination of oxygen with hemoglobin to form oxyhemoglobin

Transport of Carbon dioxide

Carbon dioxide is carried by the blood in different ways.

- As HCO_3^- ions in the plasma (about 70%): When CO_2 diffuses into the red blood cells the enzyme carbonic anhydrase catalyzes the combination of CO_2 with water to form bicarbonate (HCO_3^-) and H^+ ions. The HCO_3^- moves out of the erythrocytes into the plasma
- As carbaminohemoglobin (about 23%): CO_2 combined with protein group of hemoglobin and forms carbaminohemoglobin. Therefore CO_2 does not compete with oxygen binding sites in hemoglobin.
- Dissolved in plasma (about 7%): as free gas.

Composition and major functions of human blood

Blood which is a connective tissue is composed of cells and plasma. Cellular components of the blood are three types namely red blood cells, white blood cells and platelets. ((fig. 5.22) Red blood cells, leukocytes and platelets are developed from the bone marrow in the bones such as ribs, vertebrae, sternum and pelvis. Erythropoietin hormone (from kidneys) stimulates the generation of red blood cells.

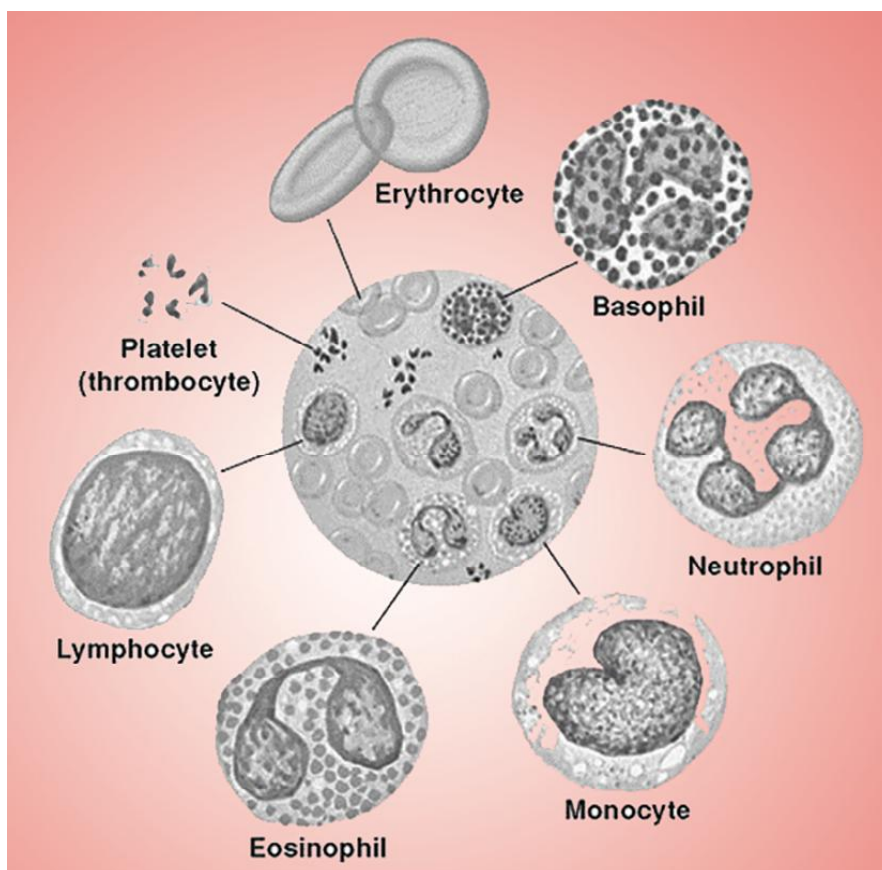


Fig 5.22: The composition of blood

Red blood cells (erythrocytes)

They are small biconcave disk-like cells. Mature erythrocytes lack nuclei. This character helps to carry more hemoglobin molecules within the cell. They also lack mitochondria thus they produce ATP via anaerobic respiration. If they produce ATP by aerobic respiration that will reduce the O_2 transport efficiency. They have about 120 days of life span.

Generally a micro liter of blood contains 4- 6 million red blood cells. This figure can be varied depending on the gender and health status.

Main function of red blood cells is the transportation of O_2 molecules. They also transport CO_2 molecules.

White blood cells (Leukocytes)

There are five types of leukocytes. They are Basophils, Lymphocytes, Eosinophils, Neutrophils and monocytes. Main functions of leukocytes are body defense, phagocytic engulfing and digesting microorganisms. Lymphocytes develop into T cells and B cells. These cells increase immune response against foreign substances.

Platelets

Platelets are derived from bone marrow cells. They also do not have nuclei and they play a major role in blood clotting.

Blood plasma

Blood plasma consists of inorganic ions in dissolved forms, plasma proteins such as albumin, antibodies and fibrinogen, nutrients, metabolic wastes, respiratory gases and hormones. pH of human blood is around 7.4. Protein concentration in plasma is higher than in interstitial fluid.

The dissolved ions in the plasma buffer and maintain the osmotic balance in the blood. Albumin in the plasma also buffers the blood and antibodies are involved in defense. Fibrinogen in the plasma aids in blood clotting. When the clotting factors are removed from the plasma it is called as serum.

Major Functions of blood

- Transport of oxygen to organs and removal of carbon dioxide from the organs and tissues
- Transport of soluble excretory materials to organs of excretion
- Transport of nutrients
- Transport of hormones from the glands where they are produced to target organs
- Defence against foreign invasions
- Aids in osmoregulation

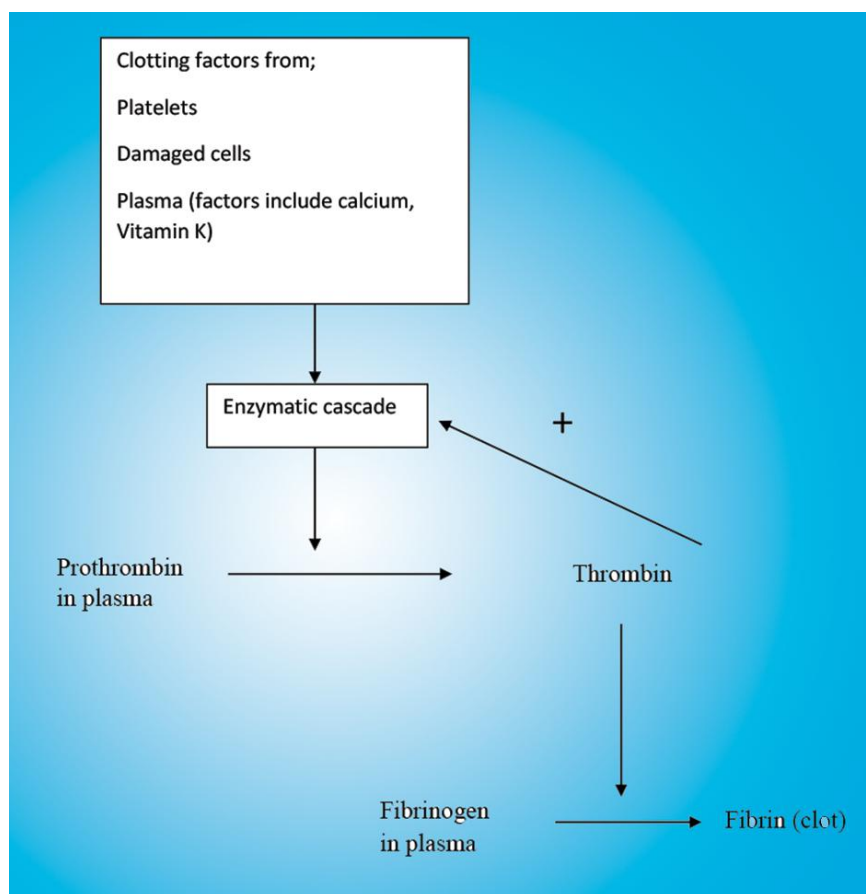
Blood clotting

When a tissue is damaged, blood flows from it and coagulates to form a blood clot. This prevents further blood loss and entry of pathogenic micro organisms which is of clear survival value. In general the blood in undamaged vessels does not clot.

A highly complex series of reactions takes place in order for coagulation to occur as well as prevent unwanted clotting. When the blood vessel is damaged the connective tissues of the vessel wall is exposed. Therefore platelets in the blood adhere to the collagen fibers in the connective tissue and release substance that makes close by platelets sticky. This platelet plug provides instant protection against blood loss.

Then platelets release clotting factors. They trigger the formation of thrombin. Then thrombin converts fibrinogen into fibrin. Next this fibrin aggregates into threads that form a network of the clot. The activated thrombin is also involved in formation of more thrombin which completes the formation of blood clot.

The cascade reaction during blood clotting is given below;



Clotting does not occur in undamaged blood vessels because the lining of the vessels is very smooth and does not promote platelet aggregation or cell rupture. Also some substances such as Heparin prevent clotting. Heparin prevent the conversion of prothrombin into thrombin and fibrinogen to fibrin and is widely used clinically as an anticoagulant.

Grouping of blood

The surface of the red blood cells carries antigens called agglutinogen (antigen A and antigen B). In addition individuals have antibodies in plasma (anti-A and anti-B). According to ABO blood grouping system there are four blood groups namely A, B, AB and O. A person with a specific antigen in red cells does not possess the corresponding antibody in the plasma. e.g. Anyone with antigen A on the red blood cell membrane has no anti-A antibody in the plasma.

If red blood cells have antigen A and plasma has antibodies b (anti-B) that person's blood group is A

If red blood cells have antigen B and plasma with antibodies a (anti-A) that person's blood group is B

If red blood cells have both antigen 'A' and 'B' and plasma has no anti-A or anti-B antibodies that person's blood group is 'AB'

If red blood cells have no antigen A or B but plasma has both antibodies (anti-A and anti-B) that person's blood group is 'O'

When a patient receives a blood transfusion it is vital that they receive blood that is compatible with their own. If it is incompatible a type of immune response occurs. This is because the donor's red cell membranes possess glycoprotein which act as antigens and react with antibodies (agglutinin) in the recipient's plasma. The result is that the donor's cells are agglutinated.

Therefore when transfusion occurs it is important to know blood group of donor and recipient. People with blood group AB make neither anti-A nor Anti-B antibodies. Transfusion of type A, B or AB blood into these individuals is likely to be safe since there are no antibodies to react with them. Person with AB blood group is known as a universal recipient.

Individual with blood group O has neither antigen A nor antigen B on their plasma membrane of red blood cells. but they do have antibodies (anti-A and anti-B) in their plasma. So these individuals having blood group O can donate blood to the persons with any blood group. A person with the blood group O is known as a universal donor. Therefore prior to transfusion cross-matching is still required to ensure that there is no reaction between donor and recipient blood.

The Rhesus system

Some individuals have antigen called Rhesus factor on the plasma membrane of red blood cells. Individuals having this factor on the red blood cells are called Rh⁺ and those who do not have this factor are called Rh⁻.

Rh⁺ individuals do not have anti Rhesus antibodies in the plasma while individuals who are Rh⁻ have anti-Rhesus antibodies in their plasma. However if Rh⁺ blood enters a Rh⁻ negative individual the recipient responds by manufacturing Rh antibodies in the blood plasma.

When a Rh⁻ mother bears a Rhesus positive (Rh⁺) child during delivery few Rh⁺ red blood cells of the fetus may enter the mother's circulation and cause the mother to produce Rh antibodies in her plasma. If the mother is pregnant for the second time with a Rhesus positive fetus, the Rh antibodies developed in her plasma in response to the first child's red blood cells can pass across the placenta to the fetus and destroy fetal red cells. Normally Rh antibodies are not formed in large enough quantities in the mother's plasma to affect the first born child. However subsequent Rh⁺ children can suffer destruction of their red blood cells.

Gas exchange in animals

Need of respiratory structures in animals and evolution of complexity of respiratory structures in different animal groups

Respiratory gas exchange (uptake of oxygen into the body and release of carbon dioxide into the external environment) occurs by diffusion. In simple animals e.g. cnidarians and flat worms every cell in the body is close enough to the external environment so that gas exchange can occur directly between all cells and the environment. Diffusion through body surface is adequate as these animals have a simple body form and low energy requirement.

In large animals, body complexity and energy requirement is high and the bulk of the cells in the body lack immediate access to the external environment. Hence, gas exchange through the body surface is not adequate to full fill their energy requirements. As a result, specialized surfaces referred to as respiratory surfaces have evolved where gaseous exchange occurs.

With the increase of the body size and complexity the surface area to volume ratio (A/V) of animals decreases. However, a large surface area is required for efficient gas exchange and thus diverse respiratory structures with large surface area with folding and branching has evolved for efficient gas exchange. Examples for such respiratory structure are Gills, Trachea, and lungs. External projections of the body such as gills were evolved in aquatic animals for efficient extraction of dissolved oxygen from water. On the other hand, surface invaginations like trachea and lungs were evolved in terrestrial animals for efficient extraction of oxygen from the atmosphere.

Characteristics of respiratory surfaces

An effective respiratory surface must have the following properties.

- It must be permeable, and wet so that gases can pass through by dissolving.
- It must be thin because diffusion is only efficient over short distances.
- It should possess a large surface area to allow a sufficient volume of gases to be exchanged according to the organism's need.
- It should possess a good blood supply (maintain a steep diffusion gradient)

Respiratory structures in animals

- Body surface: Cnidarians, Flatworms, Earthworms
- Gills